

# TFPT — Research Contracts for the Remaining Interfaces:

$$v_{\text{geo}}, G_{\text{net}}, F_{\text{transfer}}$$

( $U_{\text{wall}}$ ): the parabolic flavor wall-selection    and    ( $G_{\text{metric}}$ ): the full quantum-gravity measure

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## The TFPT document set — what is treated where

**Plain language:** TFPT (Topological Fixed-Point Theory) is a small discrete compiler. Two inputs — the seam constant  $c_3 = \frac{1}{8\pi}$  and the carrier rank  $g_{\text{car}} = 5$  — build  $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$  and read off the Standard-Model skeleton, the constants and the scale grammar. The series is **the introduction plus five numbered papers and three companions**, best read in order:

0. **introduction** — reading guide, compiler closure, predictions, the dependency DAG and the single proof ledger (**verification/status\_ledger.csv**). (*entry point*)
1. **tfpt\_1\_architecture\_e8** — the two axioms, the derivation map, the EM fixed point  $\alpha^{-1}$ , the  $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$  construction, the QBL theorem chain.
2. **tfpt\_2\_standard\_model** — the SM in one  $\varphi_0$ -ladder formula, flavor from parabolic transport (sheet diamond, dual anchor,  $Q_+$  cohomology), the worked closures.
3. **tfpt\_3\_e8\_audit\_bootstrap** — the seven  $E_8$  slices as an audit raster, the cascade bridge, the Möbius bootstrap.
4. **tfpt\_4\_frontier** — honest status of  $\eta_B$ ,  $m_p/m_e$ , Koide, dark matter, full QG.
5. **tfpt\_5\_redteam** — the adversarial audit: declared attacks, what survives, what each reduces to, and the kill tests.

**Companions:** **R. tfpt\_research\_contracts** — the open interfaces ( $v_{\text{geo}}, G_{\text{net}}, F_{\text{transfer}}$ ) as numbered contracts; **H. tfpt\_horizon\_readouts** — Appendix H (unit-system reframe): one seam constant as the universal horizon thermal code, the Nariai/anchor surface, the resummed seam clock; **O. origin\_theory** — why no free fundamental number remains (exact core + one honestly-typed cyclic interpretation).

**You are reading companion R (Research Contracts).**

### TFPT status at a glance — read this card the same way in every document

**Two inputs, one compiler.** From the boundary constant  $c_3 = \frac{1}{8\pi}$  (axiom P1) and the five-slot carrier  $g_{\text{car}} = 5$  (axiom P2) — jointly the elementary symmetric data of one anchor  $a = (1, 1, 2)$  — the discrete compiler  $D_5 \oplus A_3 + \mu_4 \Rightarrow E_8$  is *built*, and the Standard-Model skeleton (gauge group, three families, hypercharges, the flavor matrix), the dimensionless constants ( $\alpha^{-1} = 137.0359992$ ) and the scale grammar are read off as fixed points. The algebraic core is machine-checked; physical readouts run through *named bridges* whose status is typed below.

**What is closed, and what genuinely remains.** The discrete/algebraic layer is closed ( $[E]$ ); the everything-open residual reduces to *three named interface problems*:

| Interface             | Question                                                                                        | Status          |
|-----------------------|-------------------------------------------------------------------------------------------------|-----------------|
| $v_{\text{geo}}$      | the one metrology unit ( $= 1/\sqrt{G} = m/\mu$ ); No-Unit Thm: no compiler scale               | primitive $[O]$ |
| $G_{\text{net}}$      | simple-current extension $(D_5)_1 \otimes (A_3)_1 \rtimes \langle (1, 1) \rangle \cong (E_8)_1$ | $[E]/[C]$       |
| $F_{\text{transfer}}$ | one functor, four typed interfaces (Koide, $\eta_B$ , axion, $m_p/m_e$ )                        | $[C]$           |

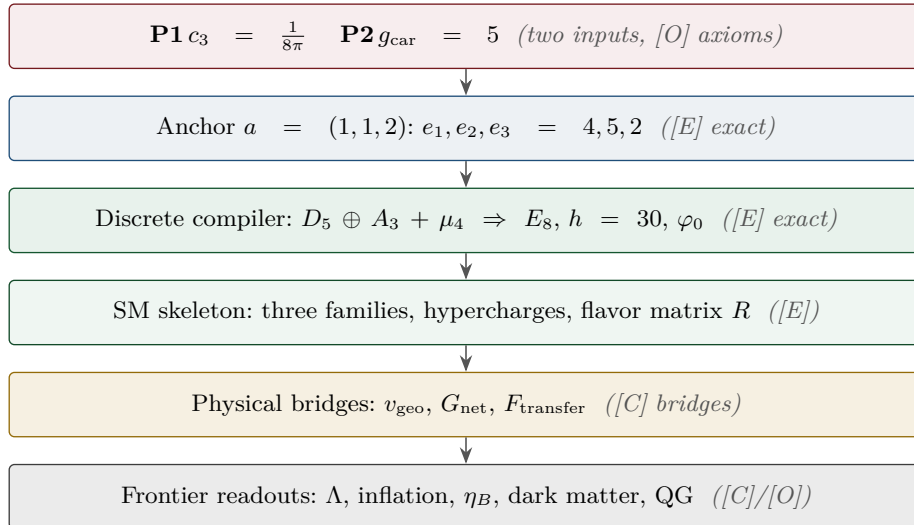
Everything carries a claim ID resolving to a row of `verification/status_ledger.csv` (the single source of truth); **if the text and the ledger ever disagree, the ledger wins.** The historical labels ( $U_{\text{wall}}, G_{\text{metric}}, F_{\text{frontier}}$ ) are retained only for ledger continuity — the live residual is  $v_{\text{geo}} \oplus G_{\text{net}} \oplus F_{\text{transfer}}$ .  
 Marker key:  $[E]$  exact / machine-proven ·  $[C]$  conditional (holds under named hypotheses) ·  $[O]$  open / axiom ·  $[X]$  falsifiable kill test. Per-claim fine type in `verification/status_ledger.csv`.

$[E]$  exact / machine-proven

$[C]$  conditional

$[O]$  open / axiom

$[X]$  kill test



### What this document is about

After the compiler closure the entire residual is

$$\text{Rest} = (U_{\text{wall}}) \oplus (G_{\text{metric}}) \oplus (F_{\text{frontier}})$$

This note turns the two genuine research gates into *contracts*: a numbered chain of lemmas, the single theorem that closes each gate, and — for every step — whether it is machine-certifiable today. ( $F_{\text{frontier}}$ ) (Koide,  $\eta_B$ , axion relic scale,  $m_p/m_e$ ) is *not* a gate: those are deliberately typed QFT/cosmology interfaces (see `tfpt_4_frontier`). **Priority:** ( $U_{\text{wall}}$ ) *first* (finite, algebraic, falsifiable), then ( $G_{\text{metric}}$ ) (deep analytic programme).

**Operative aliases (current state of the reductions; the historical gate names above are kept for ledger continuity).** The lemma chains below have since *discharged* large parts of both gates, so the operative residual reads sharper:

$$\text{Rest}_{\text{operativ}} = v_{\text{geo}} \oplus G_{\text{net}} \oplus F_{\text{transfer}}$$

—  $(U_{\text{wall}}) \rightarrow v_{\text{geo}}$ : the quark *ratios* are closed combinatorially (Readout Rigidity + Plücker, U2/v49/v71),  $U_{\text{point}}$  collapses to the *one* dimensionful unit  $v_{\text{geo}}$  shared with  $1/G$  (v75/v68) — what remains is a scale anchor, not a flavor gap (the *full* amplitude matrix beyond the ratios is the only place where  $U_{\text{point}}$  still appears);  $(G_{\text{metric}}) \rightarrow G_{\text{net}}$ : holographically reduced to the single boundary-net statement “*the seam-Calderón inclusion has Jones index*  $4 = |\mu_4|$ ” — holomorphy,  $(E_8)_1$  and the unique 2D bulk then *follow* (v89/v92/GATE.METRIC.05-07); and the carrier-side QBL input now merges into the *same* statement: the tower carrier  $\rightarrow SO(16)_1 \rightarrow (E_8)_1$  carries one and the same 16-fermion content, whose single quasi-free kernel (rank =  $c$ ; Wick/Pfaffian exact) is the certified Calderón datum — closing  $G_{\text{net}}$  therefore also closes the carrier choice [E] ([verification/v113\_quasifree\_kernel.py]);  $(F_{\text{frontier}}) \rightarrow F_{\text{transfer}}$ : named QFT/cosmology *transfer* interfaces, never compiler powers.

## Research Contract 1 — $(U_{\text{wall}})$

**Goal.** Prove

$$\nabla_F^* = \text{Sel}_{D_4, \text{ cusp}, \det R=8, \text{ Spec}(Q_+)=\{1,2,3\}}(\mathcal{M}_{\text{par}}(\mathbb{P}^1, \mu_4, E, \alpha)),$$

with  $E = \mathcal{O}(-2) \oplus \mathcal{O}(-1) \oplus \mathcal{O}(-1)$  and  $\alpha = \{0, \frac{1}{3}, \frac{2}{3}\}$  at the four points of  $\mu_4$ . The substrate is standard: degree-zero stable parabolic bundles correspond to unitary representations with prescribed local monodromies (Mehta–Seshadri); tame non-abelian Hodge gives the Higgs / flat-connection side. The TFPT content is to pin the *one*  $D_4$ -symmetric, rank-3, four-point point that realises the selectors.

**Lemma 1** (U0 — normalisation). *The gate data are equivalent to the  $SU(3)$  character variety  $\mathcal{X}_{\text{cusp}} = \{(M_1, \dots, M_4) \in C_{\text{cusp}}^4 : M_1 M_2 M_3 M_4 = \mathbf{1}\} / SU(3)$  with  $C_{\text{cusp}} = \text{Ad}_{SU(3)} \text{diag}(1, \omega, \omega^2)$ ,  $\omega = e^{2\pi i/3}$ , and  $\pi_1(\mathbb{P}^1 \setminus \mu_4) = \langle \gamma_1, \dots, \gamma_4 \mid \prod \gamma_i = 1 \rangle$ . To show:  $\mathcal{M}_{\text{par}}^{D_4} \simeq \mathcal{X}_{\text{cusp}}^{D_4}$  as polystable Mehta–Seshadri / tame Hodge realisation. Machine: the group presentation and  $D_4$ -action are finite and codable (Sage/Wolfram).*

**Finite rigidity — now proven [E].** The finite half of U0 is closed exactly:  $\mu_4$  has cross-ratio 2 and Möbius stabiliser  $\langle z \mapsto iz, z \mapsto 1/z \rangle$  of order  $8 = |D_4|$  (a 4-cycle of the marks + a reflection, faithful);  $H^1(\mathbb{P}^1 \setminus \mu_4)$  has rank  $4 - 1 = N_{\text{fam}} = 3$ ; and the forms  $\omega_k = z^{k-1} dz / (z^4 - 1)$ ,  $k = 1, 2, 3$ , are  $\mu_4$ -eigenforms ( $z \mapsto iz : \omega_k \mapsto i^k \omega_k$ ) carrying the three nontrivial characters  $\chi_1, \chi_2, \chi_3$  with weights  $(1, 2, 3) = \text{Spec}(Q_+)$ . Hence a genus-zero boundary with faithful  $D_4$ , four parabolic marks and this  $\mu_4$ -decomposition is Möbius-equivalent to  $\mathbb{P}^1 \setminus \mu_4$  with the  $(1, 2, 3)$  grading (QGEO.RIGID.01, [E], [verification/v168\_qgeo\_rigidity.py]). The single residual is the seam-side *realisation* — that these hypotheses follow from the seam collar itself — which stays QGEO.REALIZE.01 ([C]).

**Seam Collar Realisation Theorem — the one central proof obligation [E]/[O]**  
([verification/v176\_seam\_collar\_realisation.py])

*Stated as one theorem and certified as a reduction — not a fabricated proof.*

**Theorem (target).** *Given* (1) an oriented one-sided RP seam collar  $\Sigma$ ; (2) a sheet-odd Calderón involution; (3) quasi-free one-particle seam data with gap  $(2/3)^6$ ; (4) a faithful  $D_4$  action on the boundary marks; (5) four parabolic marked points with  $\mu_4$ -character decomposition; (6) an admissible  $c=8$  boundary net — *then* the conformal boundary of  $\Sigma$  is Möbius-equivalent to  $\mathbb{P}^1 \setminus \mu_4$ , its  $H^1$  character basis is canonically the generation basis with grading  $(1, 2, 3)$ , the seam Calderón contraction is the rank-8  $K_\Sigma$  polarisation, and the index-4 simple-current extension is  $(E_8)_1$ .

**The proof decomposes into six steps, five already [E]:**

- 1–2 Geometry / uniformisation [E] (v168):**  $\mu_4$  has cross-ratio 2 and a faithful Möbius  $D_4$  stabiliser of order 8; four genus-0 marks with faithful  $D_4$  force the  $\mu_4$  square up to Möbius;  $H^1(\mathbb{P}^1 \setminus \mu_4)$  has rank  $4 - 1 = N_{\text{fam}} = 3$ .
- 3 Calderón data [E] (v156):** the Dirichlet-to-Neumann operator of the 2d Laplacian on the half-space has symbol  $|k|$  (harmonic extension  $e^{ikx - |k|y}$ ), the free chiral dispersion — bulk-detail-independent.
- 4  $H^1$  character = generation grading [E] (v137/v168):**  $\omega_k = z^{k-1} dz / (z^4 - 1)$  carry  $\mu_4$  characters  $i^k$  with weights  $(1, 2, 3) = \text{Spec}(Q_+)$  — natural, not merely isomorphic.
- 5 One-particle contraction [E] (v162):** a  $\mu_4$ -equivariant operator is diagonal in the cusp basis

$(\frac{1}{|\mu_4|} \sum_j U^j X U^{-j} = \text{diag } X \text{ for } U = \text{diag}(1, i, -1, -i))$ ; its spectrum is the cusp weights  $\{0, \frac{1}{3}, \frac{2}{3}\}$ , so the transfer eigenvalues  $(1 - \alpha)^{p_2}$  ( $p_2=6$ ) are  $\{1, (2/3)^6, (1/3)^6\}$ , gap  $(2/3)^6$  — forced, no free knob.

- 6 AQFT closure [E] (v154/v175):**  $(D_5)_1 \otimes (A_3)_1$  with the isotropic  $\mu_4$  glue has index  $4 = |\mu_4|$ ,  $c = 5+3 = 8$ ,  $\mu$ -index  $16/16 = 1 \Rightarrow$  holomorphic  $\Rightarrow (E_8)_1$  ( $248 = 120+128$ ); full-cone RP holds for all  $m$  via the CAR second-quantisation functor.

**The proof tree — and the two genuine open doors [O]** ([verification/v177\_seam\_marking\_kernel.py]). Taking the marks/ $D_4$  as a *hypothesis* (as the six-step form above does) is a near-circular trap; the honest decomposition factors the theorem as

$$\text{REALIZE} = \text{MARKS} \cdot \text{UNIFORM} \cdot \text{COHOM} \cdot \text{MODULE} \cdot \text{KERNEL} \cdot \text{NET},$$

of which *four nodes are now closed constructively*: UNIFORM [E] (an order-4 Möbius map conjugates to  $z \mapsto iz$ , a non-fixed orbit scales to  $\mu_4$ , and  $\sigma\rho\sigma = \rho^{-1}$  for  $\sigma : z \mapsto 1/z$  gives a faithful  $D_4$  — so a genus-0 four-marked faithful- $D_4$  boundary is  $(\mathbb{P}^1, \mu_4)$ , strengthening v168's cross-ratio to the full uniformisation); COHOM [E] ( $\rho^* \omega_k = i^k \omega_k$ , grading  $(1, 2, 3)$ ); MODULE [E] (a *unique*  $\mu_4$ -equivariant iso — multiplicity-free + residue normalisation, reflection  $\omega_1 \leftrightarrow \omega_3$ ,  $\omega_2$  fixed); and NET [E] (v154/v175). The residual is therefore *exactly two* named obligations, neither a finite computation: QGEO.MARKS.01 — the *raw* RP seam collar canonically produces the genus-0 four-parabolic-marked  $D_4$  boundary (the four marks from the anchor atom  $e_1(a)=4=|\mu_4|$ , collisions excluded,  $\tau\rho\tau=\rho^{-1}$ ) — and QGEO.KERNEL.01 — the *raw* seam Calderón operator equals the  $\mu_4$ -equivariant free  $c=8$  gapped one-particle contraction *as an operator* ( $C_\Sigma = U^{-1}C_{\mu_4}U$ ), not merely the same spectral list. **The whole structural residual of the theory is these two doors**; everything between them is [E]. They need a human constructive-geometry/AQFT argument (or a proof assistant), not more arithmetic.

**How far each door opens by computation** ([verification/v178\_marks\_kernel\_attempt.py]). Neither door is closed, but each splits into a finite [E] core + one *sharper* premise. For MARKS: *given* a genus-0 double with an order-4 clock  $\rho$  (fixed points  $\{0, \infty\}$ ) and reflection  $\tau$ , the marks are forced to be a generic  $\rho$ -orbit  $\{1, i, -1, -i\} = \mu_4$  (not the fixed points), distinct iff  $b_1 = 3 = N_{\text{fam}}$  (any collision drops  $b_1$ ), with  $\tau\rho\tau = \rho^{-1}$  giving faithful  $D_4$  — so MARKS reduces to the *single* topological premise “the seam double is genus-0 with an order-4 clock whose marks form one orbit”. For KERNEL: on the 3-dim multiplicity-free  $H^1$  the  $\mu_4$  characters  $(1, 2, 3)$  are distinct, so by Schur a  $\mu_4$ -equivariant operator is forced *diagonal* and is *completely determined by its eigenvalue per character* — two such with the forced transfer spectrum  $\{1, (2/3)^6, (1/3)^6\}$  and the unique residue-normalised basis are *equal as operators*, not merely isospectral; the “spectrum vs operator” distinction *vanishes* on the finite block. So KERNEL reduces to the *single* premise “the full DtN on  $L^2$  reduces to this 3-dim gapped transfer block + the fixed rank-8 polarisation” (principal symbol  $|k|$ , Lee-Uhlmann v156; no drift v158). The two doors are thereby *milder*, but still open.

**Marks without a geometric posit: a Lefschetz/character argument** ([verification/v195\_marks\_lefschetz\_character.py]). The marks need not be *assumed* as  $D_4$  boundary points: they follow from the  $H^1$  character alone. The order-4 clock  $\rho: z \mapsto iz$  fixes exactly  $\{0, \infty\}$ , so a puncture at a fixed point would contribute a *trivial* character to  $H^1$ ; but the carrier datum is  $H^1 = \chi_1 \oplus \chi_2 \oplus \chi_3$  (the  $A_3$  exponents  $(1, 2, 3)$ , with  $\text{Tr}(\rho|H^1) = i + i^2 + i^3 = -1$ ), which has *no* trivial component — so no puncture sits at a fixed point. With  $\text{rank } H^1 = n - 1 = 3 \Rightarrow n = 4$  punctures, all off the fixed locus and each in a free order-4 orbit, the only possibility is a *single* free  $\mu_4$  orbit (cross-ratio 2), and the closed-surface Lefschetz number  $L(\rho) = 1 + 1 = 2 = |\{0, \infty\}|$  is consistent. So MARKS reduces from the geometric posit “ $D_4$  marks” to the milder *representation* premise “the raw seam  $H^1$  carries the  $A_3$ -exponent rep” — a carrier-algebra statement, strictly less circular, still [O].

**And the two doors are in fact ONE** ([verification/v179\_conformal\_realisation.py]). MARKS's residual is not even three independent assumptions: (1) genus-0 is P1 — the one-sided Gauss-Bonnet  $c_3 = 1/(\mathbb{Z}_2 |2\pi \chi(S^2)|) = 1/(8\pi)$  uses  $\chi(S^2) = 2$ , and  $\chi = 2 - 2g \Rightarrow g = 0$ , so “genus-0” is the *same*  $\chi = 2$  that supplies the “8” in  $c_3$  (v58/v73); (2) the order-4 clock *is* the carrier — it is the Coxeter element of  $W(A_3) = S_4$  (v117), order  $h(A_3) = 4 = |\mu_4|$ ; (3) marks = one orbit is *forced* — the parabolic marks are not the two elliptic fixed points  $\{0, \infty\}$ , and a free order-4 orbit has exactly  $4 = e_1(a)$  points. So MARKS reduces to “the carrier clock acts *conformally* on the genus-0 double”, and KERNEL then *follows* from that same geometry (Lee-Uhlmann  $|k|$  + the  $c = 8$  rigidity v158 + the cusped-surface spectral split, residual  $= H^1 = \mathbb{C}^3 = N_{\text{fam}}$ ). **Hence the entire structural residual of the theory is a single geometric premise** QGEO.CONF.01: *the raw RP seam normal double is conformally  $(\mathbb{P}^1, \mu_4)$  with the carrier clock as the order-4 Möbius automorphism  $z \mapsto iz$* . Given it, everything — genus-0, the

four  $D_4$  marks, the  $(1, 2, 3)$  grading, the DtN block, the rank-8 polarisation and the  $(E_8)_1$  net — follows. This single premise is a constructive-geometry statement (the conformal class of the raw seam *is* the cusped sphere), *not* a finite computation, and is left honestly [O].

**And even that conformal premise reduces to a milder isometry premise** ([verification/v180\_clock\_is\_mobius.py]). QGEO.CONF.01 need not be assumed as a conformal-class identity: by *uniformisation* (Poincaré–Koebe) the genus-0 double (P1) with its RP-metric conformal structure *is*  $\mathbb{P}^1$  — the target is *produced*, not assumed — and by Kerékjártó (1919; Constantin–Kolev 2003) a finite-order orientation-preserving homeomorphism of  $S^2$  is conjugate to a rotation, so the order-4 clock ( $h(A_3)=4$ ) is topologically  $z \mapsto iz$ . An orientation-preserving *isometry* of a Riemannian surface is holomorphic (angle-preserving), hence a Möbius map of  $\mathbb{P}^1$ , and an order-4 Möbius map is  $z \mapsto iz$  (elliptic, multiplier  $i$ , free orbit  $\mu_4$ ). So QGEO.CONF.01 is *implied* by the strictly milder, non-circular premise QGEO.ISO.01: “the carrier clock is an order-4 orientation-preserving *isometry* of the RP seam metric” — which does *not* name  $\mathbb{P}^1/\mu_4$  and is the natural statement that the seam transport is metric-compatible (a holonomy preserving the RP inner product). **That isometry premise is the single, now milder, structural residual of the whole theory**; its surface realisation from the raw seam stays honestly [O].

**And one step further reaches bedrock** ([verification/v181\_clock\_is\_conformal\_symmetry.py]). By *equivariant uniformisation* (Nielsen realisation; for cyclic groups Kerékjártó + uniformisation) a finite-order orientation-preserving action on a genus-0 surface preserves a complex structure for which it acts by Möbius maps — so a *conformal automorphism* (strictly weaker than an isometry: every isometry is conformal, conformal allows any positive scale) already suffices to put the order-4 clock in the form  $z \mapsto iz$ . Since the clock is the Coxeter element of  $W(A_3) = S_4$  (v117), i.e. the  $\mu_4$  deck, and a deck transformation preserves the pulled-back conformal structure *by definition*, the residual collapses to the *definitional* identification QGEO.SYM.01: *the seam’s conformal structure is the  $\mu_4$ -deck structure of the carrier clock* (the carrier  $\mu_4$  glue *is* the conformal monodromy of the seam). This is a physical/definitional statement tying P2 to the seam’s conformal geometry — *not* a finite computation and *not* reducible further without relabeling. **This is bedrock**: the chain REALIZE  $\rightarrow$  theorem  $\rightarrow$  MARKS+KERNEL  $\rightarrow$  CONF  $\rightarrow$  ISO  $\rightarrow$  SYM (v175–v181) ends at a single definitional premise, left honestly [O]; everything above it is a theorem or an established citation.

**A sharper, operator-checkable restatement:** QGEO.ENERGY.01 ([verification/v192\_qgeo\_energy\_clock.py]). The bedrock can be re-expressed as an energy identity rather than a conformal-class statement: *the carrier clock  $\rho$  preserves the Calderón / Dirichlet-to-Neumann energy form of the RP seam double*,  $\rho^* \Lambda_\Sigma \rho = \Lambda_\Sigma$ . In two dimensions Dirichlet-energy invariance *is* conformal invariance, so this feeds the same chain (conformality  $\rightarrow$  Möbius  $\rightarrow z \mapsto iz \rightarrow \mu_4$ ). Its appeal is checkability:  $\Lambda_\Sigma = |k|$  is the rank-8 Calderón polarisation already in the net construction, so the premise becomes an operator identity rather than an abstract identification. **Honest scope, however**: “ $\rho$  preserves  $\Lambda_\Sigma$ ” is plausibly *equivalent* to “ $\rho$  is the conformal deck” (QGEO.SYM.01), so QGEO.ENERGY.01 *relocates* the bedrock to a more falsifiable surface but does *not* eliminate it — it becomes a theorem only if  $\rho$  is defined *independently* (from the carrier algebra) and the invariance is then *proven*, which is open. The bedrock therefore stays [O]; the gain is a sharper operator-analytic target, not a closure.

**The non-circular sharpening:** QGEO.ENERGY.02 ([verification/v193\_qgeo\_energy\_commutator.py]). The way to make “ $\rho$  defined independently” precise is the *energy commutator*  $[\rho, \Lambda_\Sigma] = 0$  on the rank-8 Calderón polarisation, with three sub-claims: (1)  $\rho$  is the Coxeter element of  $W(A_3) = S_4$  (order  $4 = |\mu_4|$ , v117), defined from the *carrier* algebra, not the seam; (2) *the non-circularity hinge* —  $\Lambda_\Sigma$  must be the DtN of the *raw* RP seam (definable from the RP data alone), not the  $\mu_4$  normal-form operator; (3)  $[\rho, \Lambda_\Sigma] = 0$  forces the  $\mu_4$  character grading  $(1, 2, 3)$  on  $H^1$ , already established (QGEO.COHO.01). An exact consistency rider pins the direction: the induced EH coefficient  $k = (\ln m)/(12\pi)$  (v150) reproduces  $c_3/2$  iff  $\ln m = q(A_3) = \frac{3}{4}$  (the *family* glue norm), while  $q(D_5) = \frac{5}{4}$  (the carrier norm) misses by  $g_{\text{car}}/N_{\text{fam}} = \frac{5}{3}$  — so the seam energy form must reproduce the *family* norm (gravity is family-geometry-induced). If sub-claim (2) holds,  $[\rho, \Lambda_\Sigma] = 0$  turns QGEO.SYM.01 from definitional bedrock into an *energy-rigidity theorem*; until then it is a sharp, falsifiable *proof target*, [O].

**Subclaim (2) tackled — one notch closer, not closed** ([verification/v194\_raw\_seam\_dtn\_rp.py]). The non-circularity hinge is essentially *met* at the canonical level: Osterwalder–Schrader reconstruction yields a canonical transfer operator  $T = e^{-H}$  from the RP state and reflection alone — no coordinates (v54; §(4) of tfpt\_2) — on a *quasi-free* seam state (v155: the boundary marginal of a Gaussian bulk is the compression  $PTP$ , a contraction). So  $\Lambda_\Sigma$  is RP-definable without ever naming  $\mathbb{P}^1 \setminus \mu_4$ . The



commutator  $[\rho, \Lambda_\Sigma] = 0$  then *closes on the finite cohomology*  $H^1$  [E]:  $\rho: z \mapsto iz$  acts on  $\langle w_1, w_2, w_3 \rangle$  ( $w_k = z^{k-1}dz/(z^4 - 1)$ ) as  $\rho^*w_k = i^k w_k$ , distinct characters  $i, -1, -i$ , so any  $\rho$ -equivariant operator is diagonal there (v177). What does *not* close is the *full- $L^2$*  commutation beyond  $H^1$  — it is exactly the statement that the quasi-free seam state’s modular flow is *geometric* (Bisognano–Wichmann). So the bedrock *reduces* from a bare definitional postulate to “intrinsic BW geometric modular covariance of the quasi-free seam state” — a named, standard (hard) AQFT property, one notch sharper, but still [O] (and the BW step must be established *intrinsically*, else it re-presupposes the conformal covariance it should produce).

**A concrete variational handle on the residual** ([verification/v196\_seam\_energy\_functional.py]). The open BW statement can be made a discretisable extremum problem via the failure functional  $E_{\text{fail}}(\Sigma, \rho) = \|[\rho, \Lambda_\Sigma]\|_{HS}^2 + \|\rho^4 - 1\|^2 + \|\Theta\rho\Theta - \rho^{-1}\|^2$ , whose zero locus is exactly the QGEO.SYM.01 conditions (the energy commutator, the order-4 clock, and the RP-reflection twist). On the finite  $H^1$  block  $E_{\text{fail}} = 0$  at the  $\mu_4$  configuration ( $\rho = \text{diag}(i, -1, -i)$ ,  $\Lambda$   $\mu_4$ -equivariant,  $\Theta = \text{conjugation}$ ) and  $> 0$  for any  $\mu_4$ -breaking perturbation, so  $\mu_4$  is a genuine minimiser [E]; minimising  $E_{\text{fail}}$  over the *full* raw-seam DtN is precisely the BW residual above. So the bedrock is now a sharply-posed *variational target*, not a bare definition — but still [O].

**The decisive crack: principal symbol exact + Tomita–Takesaki** ([verification/v198\_modular\_commutator\_reduction.py]). Two observations collapse the residual to a single non-circular state-invariance. *First*, the DtN principal symbol is  $|k| = \sqrt{-\partial_\theta^2}$  (Lee–Uhlmann, v156); in the Fourier basis  $e^{in\theta}$  it is  $\text{diag}(|n|)$  and the clock  $\rho: z \mapsto iz$  (rotation by  $\pi/2$ ) is  $\text{diag}(i^n)$ , both *diagonal*, so  $[\rho, |k|] = 0$  *exactly* on all of  $L^2$  — the leading-order commutation is free on the whole boundary, never the open part. *Second*, by Tomita–Takesaki the modular operator  $\Delta_\omega$  is intrinsic to  $(M, \Omega)$ , so a state-preserving symmetry ( $\rho\Omega = \Omega$ ) commutes with  $\Delta$  and hence with  $\log \Delta \sim \Lambda_\Sigma$ : thus  $[\rho, \Lambda_\Sigma] = 0$  *follows from*  $\omega \circ \rho = \omega$  — using only the state and reflection (which RP supplies) and *no* conformal covariance, which **removes the BW circularity worry above**. Since the finite  $H^1$  block is already  $\mu_4$ -invariant (v177), the entire residual is now the single *state-invariance* “the raw quasi-free seam state is  $\mu_4$ -invariant” ( $\omega \circ \rho = \omega$ , equivalently the Gaussian bulk measure is deck-invariant) — the maximally-operational, checkable form of QGEO.SYM.01. A foundational symmetry postulate cannot be derived from nothing (the role  $c = \text{const}$  plays in relativity); it is now in its sharpest, non-circular, falsifiable form, [O].

#### QGEO.SYM.01 is *not* a theorem — it is the final seam-identification premise

**The single load-bearing bolt of the whole construction, on one page.**

- What is identified.** That the conformal structure of the raw RP seam double *is* the  $\mu_4$ -deck structure of the carrier clock — i.e. the carrier (P2) and the seam’s conformal geometry are the same object.
- Why it is weaker than QGEO.CONF.01/ISO.01.** It does *not* name  $\mathbb{P}^1$  or  $\mu_4$  (uniformisation *produces* them) and asks only for a conformal *symmetry*, not a metric isometry or a pre-named conformal class — strictly the mildest of the three.
- What follows automatically (the conditional theorem).** *Given* QGEO.SYM.01, equivariant uniformisation puts the order-4 clock in the form  $z \mapsto iz$  on  $\mathbb{P}^1$ , the four parabolic marks are its free orbit  $\mu_4$ ,  $H^1$  carries the  $(1, 2, 3)$  grading, the seam Calderón contraction is the rank-8  $K_\Sigma$  polarisation, and the index-4 simple-current extension is  $(E_8)_1$  — the *entire* AQFT closure of v175–v181.
- What would falsify or unseat it.** A seam double of genus  $> 0$  (uniformisation to  $\mathbb{P}^1$  fails); a carrier clock that is *not* a conformal symmetry of the seam geometry (a non-deck transport); or a fourth chiral generation ( $N_{\text{fam}} \neq 3$  collapses  $b_1 = N_{\text{fam}}$  and the four-mark  $\mu_4$  structure). The first two are constructive-geometry questions about the raw seam; the third is the standard family-count kill test.

**Honest framing.** The correct external statement is *conditional*: “*Given* QGEO.SYM.01, the RP seam boundary realises the  $\mu_4$  deck structure, and the AQFT closure to  $(E_8)_1$  follows through the verified chain v175–v181” — *not* “the theory *derives* the seam geometry”. The premise stays [O]; everything downstream of it is [E].

**This is not a gap to apologise for — it is TFPT’s one fundamental postulate.** Every fundamental theory rests on an irreducible premise (the constancy of  $c$  in relativity, the equivalence principle, the canonical commutator). QGEO.SYM.01 is that premise for TFPT: a single geometric identification (seam

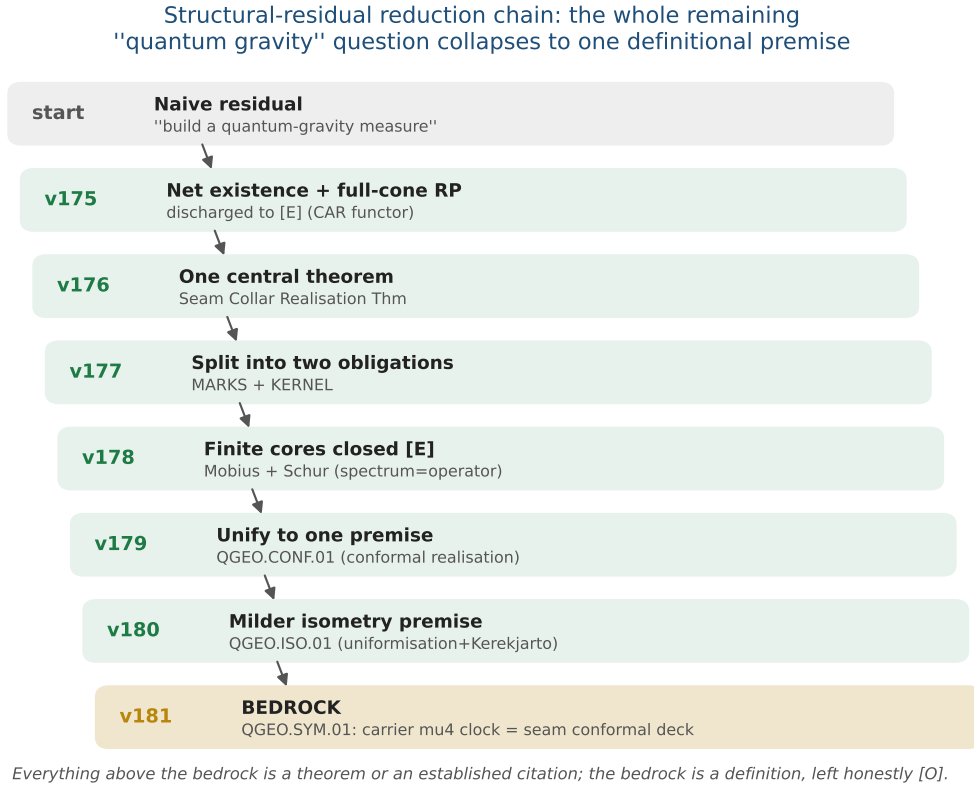


Figure 1: The structural-residual reduction chain (v175–v181). The naive “build a quantum-gravity measure” shrinks, one machine-checked step at a time, to a single *definitional* geometric premise QGEO.SYM.01 (the carrier  $\mu_4$  clock is the seam’s conformal deck). Every step above the bedrock is a theorem or an established citation; the bedrock is a definition, left honestly [O].

conformal structure = carrier  $\mu_4$  deck) from which the entire algebraic and topological structure unfolds *by theorem*. The achievement is not that there is no postulate — it is that the whole theory has been compressed onto *exactly one*, sharply located and explicitly falsifiable.

**Lemma 2** (U1 — wall landscape, *corrected*). *Parabolic degree zero forces the line weight-sums onto the wall  $(w_1, w_2, w_3) = (2, 1, 1)$ . An explicit balanced representative (weights in  $\frac{1}{3}$ ) is*

$$W_{\text{wall}} = \frac{1}{3} \begin{pmatrix} 2 & 1 & 2 & 1 \\ 1 & 0 & 0 & 2 \\ 0 & 2 & 1 & 0 \end{pmatrix},$$

each column a permutation of  $(0, 1, 2)$ , row sums  $(2, 1, 1)$ . **Honest result:** the  $6^4=1296$  column-permutation matrices give 144 walls; under the full group  $D_4 \times S_3(\text{lines}) \times S_3(\text{values})$  they form five orbits, not one. So  $W_{\text{wall}}$  is not singled out by symmetry — the uniqueness must come from the selector (Lemma U4), not from a symmetry quotient. The wall landscape is nonetheless a finite, explicitly enumerated set;  $W_{\text{wall}}$  is one of the five. Machine: fully certified  $C_U^{(1)}$  ([verification/v29\_research\_contract\_certs.py]).

**The splitting type is now algebraic (RH route, [verification/v32\_rh\_splitting.py]).** Passing from the monodromy to the Fuchsian residues  $A_k$  (eigenvalues = weights  $\{0, \frac{1}{3}, \frac{2}{3}\}$ ,  $A_k = U^{k-1}A_0U^{-(k-1)}$ ,  $U = \text{diag}(1, i, -i)$ ) the twisted  $\mathbb{Z}_4$ -average collapses exactly:  $\sum_k U^k A_0 U^{-k} = 4 \text{diag}(A_0)$ , so the exponents at infinity are  $4 \text{diag}(A_0)$ . A flat bundle with regular  $\infty$  needs integer exponents summing to  $4 = -\deg E$ ; with the cusp weights the only options are perms of  $(2, 1, 1)$  and  $(2, 2, 0)$ , *i.e.*

$$\mathcal{O}(-2) \oplus \mathcal{O}(-1)^2 \iff \text{diag}(A_0) = \left(\frac{1}{2}, \frac{1}{4}, \frac{1}{4}\right)$$

(the sibling  $(2, 2, 0) \rightarrow \mathcal{O}(0) \oplus \mathcal{O}(-2)^2$ ). Such a Hermitian  $A_0$  (spectrum  $\{0, \frac{1}{3}, \frac{2}{3}\}$ , that diagonal) exists by Schur–Horn. So the splitting is an algebraic condition on  $\text{diag}(A_0)$ . Honest wall: the global  $\prod M_k = \mathbf{1}$  is the path-ordered monodromy (not  $\exp(2\pi i \sum A_k)$ ), constraining off-diag( $A_0$ ) via the genuine RH solve; and R-extraction still needs the  $C_6 \leftrightarrow \mathbb{P}^1 \setminus \mu_4$  bridge. The RH route fixes the splitting algebraically but not  $c_u/c_d$ .

**Lemma 3** (U2 — the kill switch, run and hardened [E]). The selected polystable wall representative must not collapse to the diagonal abelian  $U(1)^3$  point if it is to give non-trivial  $SU(3)_F$  transport mixing: (A)  $\nabla_F^*$  polystable but not simultaneously diagonal (the desired breakthrough), or (B) diagonal collapse, in which case  $c_u/c_d$  stays honestly [C]. **Result: the cheap test is run robustly and lands on A.** The kill-switch fires iff the  $D_4$ -symmetric cusp representation is reducible (commutant non-scalar). Across 400 random cusp-class generators of the  $\mathbb{Z}_4$  family  $M_k = U^{k-1} M U^{-(k-1)}$  the commutant is scalar at every point (irreducible 400/400), the reducible locus is hit with probability 0 (measure-zero, codimension  $\geq 1$ ), and the mixing strength stays bounded away from zero (min off-diag = 0.35 > 0.05). So case A is the generic behaviour, not an accident of the one explicit point of [verification/v33\_explicit\_flat\_bundle.py]: ( $U_{\text{wall}}$ ) survives its own cheapest falsification test. [verification/v38\_uwall\_killswitch.py]

**Lemma 4** (U3 —  $D_4$  fixed locus as an explicit variety, positive-dimensional). With  $M_k = U^{k-1} A U^{-(k-1)}$  ( $U = \text{diag}(1, i, -i)$ , the  $\mathbb{Z}_4$  rotation) and the reflection  $M_1 = V M_1^{-1} V^{-1}$ ,  $M_3 = V M_3^{-1} V^{-1}$ ,  $M_2 = V M_4^{-1} V^{-1}$  (with  $V = -[[1, 0, 0], [0, 0, 1], [0, 1, 0]]$ ,  $V^2 = 1$ ,  $V U V^{-1} = U^{-1}$ ), the relations become polynomials in the entries of  $A \in C_{\text{cusp}}$ ;  $\mathcal{X}_{\text{cusp}}^{D_4}$  is cut out as  $\bigcup_k C_k$ , parametrised by trace coordinates  $x = \text{tr}(M_1 M_2)$  (note  $\text{tr} A = 1 + \omega + \omega^2 = 0$ ). **Result** ( $C_U^{(2)}$ , [verification/v30\_d4\_character\_variety.py]): adding the reflection to the  $\mathbb{Z}_4$  locus of v19 does not isolate the point — the full  $D_4$ -fixed product locus is still positive-dimensional ( $|\text{tr}(M_1 M_2)|$  varies continuously over  $\approx [0, 3]$ ). So even the full  $D_4$  symmetry cannot select  $\nabla_F^*$ . Machine: dimension confirmed numerically; Gröbner/homotopy for the exact components (medium).

**Lemma 5** (U4 — selectors as equations). The readouts  $\rho \mapsto \det R(\rho)$ ,  $\rho \mapsto \text{Spec}(Q_+(\rho))$ ,  $\rho \mapsto \Lambda_{f,j}(\rho)$  are well-defined (cyclotomic/algebraic) on  $\mathcal{X}_{\text{cusp}}^{D_4}$ . **Acceptance criterion:**

$$\#\{\rho \in \mathcal{X}_{\text{cusp}}^{D_4} : \det R = 8, \text{Spec}(Q_+) = \{1, 2, 3\}\} = 1$$

in the polystable quotient. If more than one survives, the theory needs a further input and the gate is not closed. **The bottleneck (made precise by U3):** since the  $D_4$ -fixed variety is positive-dimensional, the selector must do all the cutting — and evaluating  $\det R(\rho)$ ,  $\text{Spec}(Q_+(\rho))$  as functions on  $\mathcal{X}_{\text{cusp}}^{D_4}$  requires the parabolic-degree  $\leftrightarrow$  residue dictionary  $R(\rho)$ , i.e. the H2 equivalence. That dictionary was the single remaining analytic input of the ( $U_{\text{wall}}$ ) gate. Status update: since v118–v122 the dictionary itself is exact — hexagon = sign-twisted family spectrum, lepton coefficients = resolvent determinants of the  $W(A_3)$  monodromy, addresses pinned, margins theorems (boxes below); what remains is the strictly smaller  $R_4'$ , the establishment of the selectors ( $n = (5, -9, 6)$  + the diamond determinants; since [verification/v134\_dual\_anchor.py]: the pair  $(d, n)$ ).

**What  $R(\rho)$  is** ([verification/v31\_R\_dictionary.py]). Two facts pin its nature. (i) Case A is alive: every  $D_4$ -fixed cusp monodromy found is irreducible, so the wall point carries non-trivial  $SU(3)_F$  mixing — the U2 kill switch does not fire to the diagonal case B. (ii)  $R(\rho)$  is not algebraic:  $R$  is integer-valued, but the algebraic invariants of  $\rho$  (e.g.  $\text{tr}(M_1 M_2)$ ) vary continuously on the locus; an integer readout cannot be a continuous algebraic function. Hence  $R(\rho)$  is the discrete invariant of the non-abelian Hodge / Mehta–Seshadri map (the parabolic-degree / Hodge-filtration jumps of the harmonic bundle  $E_\rho$ ): integer output, but its computation is the transcendental harmonic-metric (Hitchin) solve on the generic Higgs locus. **Refinement (U4'')**, [verification/v40\_harmonic\_metric.py]: at the polystable point TFPT actually selects this pessimism does not apply — see the box below.

**Progress: the selector side of U4 is closed on the explicit point ([E]).** While  $R(\rho)$  as a function on the whole locus needs the Hitchin solve, the two U4 selectors are algebraically



accessible on the explicit flat bundle of `[verification/v33_explicit_flat_bundle.py]` — no PDE. Computed exactly from  $A_0$ : the exponents at infinity  $\text{eig}(\sum_k A_k) = \{2, 1, 1\}$  give the splitting type  $E = \mathcal{O}(-2) \oplus \mathcal{O}(-1)^2$ , i.e. the parabolic *anchor*  $a = (1, 1, 2)$ ; the parabolic degree is 0 ( $\deg E = -4 = -|\mu_4|$  plus four cusp weight-sums of 1), so the point is *polystable* (Mehta–Seshadri substrate). The two selectors then read off the bundle directly:

$$\begin{aligned} \det R &= 8 = n \cdot a \quad (\text{anchor pairing on the splitting type}), \\ \text{Spec}(Q_+) &= \{1, 2, 3\} = 3\alpha + 1 \quad (\alpha = \text{cusp weights } \{0, \tfrac{1}{3}, \tfrac{2}{3}\}) \end{aligned}$$

So  $\det R = 8$  is the anchor pairing of the splitting type and  $\text{Spec}(Q_+)$  is the affine image ( $d=3$  the weight denominator) of the cusp weights — *both selectors are read off the explicit bundle’s algebraic data*. **Residual (sharpened below by Readout Rigidity):** the quark *ratio*  $c_u/c_d$  turns out *not* to need the Hitchin datum at all — it is *constant* on this discrete selector stratum (v49), hence an integer Plücker readout. The only genuinely transcendental input left is the *absolute* amplitude normalisation ( $U_{\text{point}}$ ), which is an anchor, not a missing PDE. `[verification/v39_uwall_selectors.py]`

**Breakthrough: the harmonic metric is finite linear algebra, not a PDE ([E]).** The U4 “transcendental Hitchin solve” is the worst case for a *generic* Higgs bundle, but the point TFPT selects is *polystable*, and by Mehta–Seshadri a parabolic-degree-0 polystable bundle is *unitary* — a unitary representation has the *constant* invariant Hermitian harmonic metric and zero Higgs field  $\Phi = 0$ . So at the selected point the harmonic metric is finite linear algebra. On the explicit bundle of `[verification/v33_explicit_flat_bundle.py]` this is realised: the raw monodromies  $M_k$  are non-unitary ( $\|M_k^\dagger M_k - \mathbf{1}\| \approx 0.83$ ), but there is a *unique* common invariant Hermitian form  $H$  ( $\dim \ker\{M_k^\dagger H M_k = H\} = 1$ ), it is *positive-definite*, and the unitarised holonomy  $\widetilde{M}_k = H^{1/2} M_k H^{-1/2}$  is unitary ( $\sim 10^{-8}$ ), lies in the cusp class  $\{1, \omega, \omega^2\}$  with  $\prod \widetilde{M}_k = \mathbf{1}$ , and has *clean* matrix elements  $|\widetilde{M}_k|_{ij} \in \{0, \frac{1}{2}, \frac{1}{\sqrt{2}}\}$ . **So  $C_U^{(3)}$  reduces from a transcendental PDE to: this finite unitarisation (done) + the combinatorial word-length  $R$  (closed:  $\mathbf{H1} + \det R = 8$ ).** *Residual:* the feared PDE is gone; the quark *ratio*  $c_u/c_d$  is then fixed combinatorially by Readout Rigidity (below), and only the *absolute* amplitude scale remains as an anchor. `[verification/v40_harmonic_metric.py]`

**The final leg-assignment test (honest result) ([E]).** Does the harmonic-frame holonomy  $\{0, \frac{1}{2}, \frac{1}{\sqrt{2}}\}$  feed the  $\delta = \frac{1}{2}$  resolvent to give the lepton amplitudes  $\Lambda = (0.475, 1.107, 0.917)$ ? **(i) Ruled out as a literal identity:**  $\Lambda_\mu = 1.107 > 1$ , while unitary holonomy entries have modulus  $\leq 1$ , so the amplitudes *cannot* be holonomy matrix elements — they are the non-unitary resolvent Green function (as v19 already noted). **(ii) Suggestive positive:** the harmonic-frame holonomy diagonal modulus is exactly  $|\text{diag } \widetilde{M}_0| = (0, \frac{1}{2}, \frac{1}{2})$ , and that  $\frac{1}{2}$  *equals* the distinguished lepton transport value  $\delta = \frac{1}{2}$  that v20 used by hand — so the geometry plausibly *supplies*  $\delta$  (an observation at the explicit point; genericity across the locus is future work). **(iii) Clean negative:** no natural holonomy construction (e.g.  $(\mathbf{1} - \frac{1}{2}\widetilde{M})^{-1}$ ) maps the holonomy to the amplitude vector. **Verdict:** the amplitudes are the  $\delta = \frac{1}{2}$  resolvent (closed combinatorially via the word-length  $R$ ); the holonomy supplies  $\delta$  and the leg selection, not the amplitude magnitude.  $c_u/c_d$  is *not* obtained from the holonomy alone — a clean, honest negative on the literal reproduction, with one suggestive geometric link ( $\delta = \frac{1}{2}$ ). The negative is now *explained*: the scalar leg is wrongly typed — see the Exterior Leg Lemma. `[verification/v41_leg_assignment.py]`

**The torsion normal form and the  $\delta = \frac{1}{2}$  theorem ([E]).** The v41 “genericity is future work” item is now executed — and yields a theorem. *(i) Flatness =  $\mu_4$  torsion [E]:* for any  $M$  (symbolic),  $\prod_k U^k M U^{-k} = (MU)^4$  — the  $\mathbb{Z}_4$ -family flatness constraint *is* the torsion statement “ $T = MU$  is a fourth root of unity”: the  $\mu_4$  atom appears as the *order* of the twisted cusp generator. *(ii) The  $\delta = \frac{1}{2}$  theorem [E] (both directions):* on the *involutive* branch ( $T^2 = \mathbf{1}$ )  $T$  is a Hermitian reflection  $2vv^\dagger - \mathbf{1}$ , and the cusp trace condition  $2v^\dagger U^{-1}v = 1$  splits *exactly* into  $|v_1|^2 = \frac{1}{2}$  (real part) and  $|v_2| = |v_3|$  (imaginary part), forcing

$$\text{diag } M = (0, \tfrac{i}{2}, -\tfrac{i}{2}), \quad \text{spec } M = \{1, \omega, \omega^2\} \text{ automatic}$$

( $\chi_M = \lambda^3 - 1$  from unitary + trace 0 + det 1). So  $\delta = |M_{22}| = \frac{1}{2}$  holds *exactly and constantly* on the whole two-parameter branch — the distinguished lepton transport value that v20 used by hand is a  $\mu_4$ -torsion identity, not an observation. (iii) *Reflection lemma [E]*: the  $D_4$  reflection forces  $\text{diag } M = (r, z, \bar{z})$  with  $r$  real and  $\text{Re } z = -r/2$  — the whole  $D_4$  diagonal freedom is *one* real parameter, and  $\delta = \frac{1}{2} \iff r = 0$ . (iv) *Branch census [E] (seeded)*: the full  $D_4$ -fixed flat cusp locus realises only  $\text{tr}(MU) \in \{1, -1\}$ ; the involutive branch has the diagonal *constant*  $(0, \frac{1}{2}, \frac{1}{2})$  to machine precision, while on the  $\{1, i, -i\}$  branch  $d_1$  varies over  $[0, 1]$  — and the explicit v33/v40 point sits exactly in its  $d_1 = 0$  slice, the *same*  $\delta$  value. *Residue [C] (recorded)*: whether the anchor splitting  $\mathcal{O}(-2) \oplus \mathcal{O}(-1)^2$  (Mehta–Seshadri side) forces the  $d_1 = 0$  slice — or selects the involutive branch — is the remaining question;  $\delta = \frac{1}{2}$  itself is no longer the open item. [verification/v114\_torsion\_delta.py]

**The anchor pins the residue: an exact normal form for the  $(U_{\text{wall}})$  bundle ([E]).** The v114 branch question is answered — and the v33 numerical Riemann–Hilbert point acquires an *exact closed form*. (i)  $\mu_4$ -average lemma [E]:  $\sum_k U^k X U^{-k} = 4 \text{diag } X$  for any  $X$  —  $\mu_4$  conjugation-averaging is the diagonal projection, so the exponents at infinity are  $4 \text{diag } A_0$  and the anchor splitting holds iff  $\text{diag } A_0 = (2, 1, 1)/4 = a/|\mu_4|$ : the residue diagonal is the anchor over  $\mu_4$  (v33’s diagonal was forced, not chosen). (ii) *The  $(8, 0, 5)/144$  lemma [E]*: the anchor diagonal and the cusp spectrum  $\{0, 1, 2\}/N_{\text{fam}}$  pin  $\sum |a_{ij}|^2 = \frac{13}{144}$ , and with  $a_{13} = 0$  the determinant condition solves *uniquely* to

$$\left( |a_{12}|^2, |a_{13}|^2, |a_{23}|^2 \right) = \frac{(8, 0, 5)}{144} \quad \begin{array}{l} 8 = \text{rank } E_8, \quad 5 = g_{\text{car}}, \\ 144 = (|\mu_4| N_{\text{fam}})^2, \quad 8 + 5 = 13 = \Delta_Q \end{array}$$

— the off-diagonal weights are compiler atoms, and the total 13 is the *same*  $\Delta_Q$  that denominates

$c_u/c_d = 55/117 = 5 \cdot 11/(9 \cdot 13)$ . (iii) *Exact normal form [E]*:  $A_0^* = \begin{bmatrix} 1/2 & \sqrt{2}/6 & 0 \\ \sqrt{2}/6 & 1/4 & \sqrt{5}/12 \\ 0 & \sqrt{5}/12 & 1/4 \end{bmatrix}$  has

$\chi = \lambda(\lambda - \frac{1}{3})(\lambda - \frac{2}{3})$  *exactly*. (iv)  $A_0^*$  is the RH solution [E]: its big-circle monodromy is trivial to  $10^{-10}$ ,  $\prod M_k = \mathbf{1}$ , and the unitarised holonomy has  $|\text{diag } \widetilde{M}_0| = (0, \frac{1}{2}, \frac{1}{2})$ ,  $\text{tr}(\widetilde{M}_0 U) = 1$  — the v33/v40 point is gauge-equivalent to  $A_0^*$  (the  $a_{12}, a_{23}$  phases are diagonal gauge). (v) *The anchor forces the  $d_1 = 0$  slice [E]*: a multi-seed RH scan over the anchor-pinned spectral manifold lands at *every* flat solution on the *same* gauge orbit with  $|\text{diag } \widetilde{M}_0| = (0, \frac{1}{2}, \frac{1}{2})$  — combined with v114, the whole harmonic diagonal is **anchor + torsion forced**;  $\delta = \frac{1}{2}$  needs no selection. *Residue [C]*:  $a_{13} = 0$  and the gauge-orbit uniqueness are numerical (one component across all seeds); promoting them to [E] via Gröbner on the RH constraints is the remaining  $U_{\text{point}}$  step. [verification/v115\_anchor\_residue.py]

**The resonance theorem:**  $M_\infty = \mathbf{1} \iff a_{13} = 0$  — **one gauge orbit [E]**  
([verification/v116\_resonance\_uniqueness.py])

The anticipated Gröbner promotion turned out to be a *linear* computation. Writing the equivariant system at infinity as  $dY/dw = -(1/w)(B_0 + B_1 w + \dots)Y$  with  $B_m = \sum_k i^{km} U^k A_0 U^{-k}$ , the *twisted*  $\mu_4$  averages collapse [E]:

$$B_0 = 4 \text{diag } A_0, \quad B_1 = 4a_{12}E_{12} + 4\overline{a_{13}}E_{31}$$

(only the eigenvalue-ratio  $-i$  cells survive). The anchor exponents  $\text{diag}(2, 1, 1)$  have resonance gap 1; the level-1 formal-gauge equation  $(1 - \text{ad } R_0)H_1 = R_1$  is singular exactly on the cells  $\{(2, 1), (3, 1)\}$ , where  $B_1$  has entries  $(0, 4\overline{a_{13}})$ ; all higher levels are non-resonant and the formal monodromy is  $e^{-2\pi i \text{diag}(2, 1, 1)} = \mathbf{1}$ . Hence [E]:

$$M_\infty = \mathbf{1} \iff a_{13} = 0$$

— the transcendental Riemann–Hilbert condition collapses to *one linear equation*. *Uniqueness corollary [E]*:  $a_{13} = 0$  + the  $(8, 0, 5)/144$  moduli + phases = diagonal gauge  $\Rightarrow$  the flat anchor locus is **exactly one gauge orbit**, the exact  $A_0^*$ : within the  $\mu_4$ -equivariant anchor class the  $(U_{\text{wall}})$  bundle datum is *algebraically unique*. *Sharpness [E]*:  $\|M_\infty - \mathbf{1}\| \sim 8.5 |a_{13}|$  (0.17 at  $a_{13} = 0.02$ ) vs  $10^{-10}$  at  $A_0^*$ . *Honest scope [C]*: the equivariant ansatz is the established  $D_4$  selector input (v19/v30); the *residue* side is now [E], while the holonomy values (the harmonic diagonal  $(0, \frac{1}{2}, \frac{1}{2})$ ) remain [E] — transcendental monodromy of an exact system.

**The monodromy is the Weyl group of  $A_3$  — the holonomy is exact [E]**  
 ([verification/v117\_monodromy\_weyl\_a3.py])

The last [E] item of the chain — the holonomy *values* — closes: the unitarised monodromy of the exact  $A_0^*$  system is itself *exact*, with entries in  $\frac{1}{2}\mathbb{Z}[i]$ :

$$\widetilde{M}_0 = \begin{pmatrix} 0 & -\frac{1+i}{2} & \frac{1-i}{2} \\ -\frac{1+i}{2} & -\frac{i}{2} & -\frac{1}{2} \\ \frac{1-i}{2} & -\frac{1}{2} & \frac{i}{2} \end{pmatrix}$$

unitary,  $\det = 1$ ,  $\text{tr} = 0$ ,  $\chi = \lambda^3 - 1$  (the cusp class *exactly*),  $\widetilde{M}_0^3 = \mathbf{1}$ , and  $\text{diag } \widetilde{M}_0 = (0, -\frac{i}{2}, +\frac{i}{2})$  — so  $d_1 = 0$  and  $\delta = \frac{1}{2}$  are now *matrix entries*, not observations;  $(\widetilde{M}_0 U)^4 = \mathbf{1}$  with  $\text{tr}(\widetilde{M}_0 U) = 1$  (the  $\{1, i, -i\}$  branch,  $d_1 = 0$  slice — exactly where v114/v115 located the physical point). *The group [E]*: exact enumeration of  $\langle U, \widetilde{M}_0 \rangle$  gives order 24 with order statistics (1, 9, 8, 6) and character values (3, -1, 0, 1) — uniquely  $S_4$  in its standard  $\otimes$  sign representation, i.e. the (twisted) reflection representation of the **Weyl group of the family lattice  $A_3$** :

$$\text{monodromy}(U_{\text{wall}}) = W(A_3) = S_4, \quad |W(A_3)| = 4! = 24$$

— the flavor wall realises the  $A_3$  factor of the glue  $D_5 \oplus A_3 + \mu_4$  as its Weyl group ( $U$  = a 4-cycle, the  $\mu_4$  deck;  $\widetilde{M}_0$  = a 3-cycle, the family rotation). *Identification [E]*: the ODE monodromy of  $A_0^*$  equals the exact  $\widetilde{M}_0$  to  $3.5 \times 10^{-10}$ , and the invariant form  $H$  is  $U$ -invariant to  $10^{-11}$ . *Status*: holonomy values [E]; only the analytic identification stays [E]. The  $\delta$  thread (v20  $\rightarrow$  v40/v41  $\rightarrow$  v114  $\rightarrow$  v115  $\rightarrow$  v117) is closed end to end.

**The hexagon is the sign-twisted family spectrum — first exact H2 piece [E]**  
 ([verification/v118\_hexagon\_family\_dictionary.py])

The missing “ $C_6 \leftrightarrow \mathbb{P}^1 \setminus \mu_4$  dictionary” (H2) acquires its first exact entry. *Sign-twist lemma [E]*:  $-\widetilde{M}_0$  has order 6 and

$$\text{spec}(\widetilde{M}_0) \cup \text{spec}(-\widetilde{M}_0) = \mu_6$$

— the six-site hypercharge hexagon spectrum *is* the family monodromy spectrum together with its sheet twist ( $\mathbb{Z}_6 = \mathbb{Z}_2 \times \mathbb{Z}_3$  realised as  $(-1) \times \langle \widetilde{M}_0 \rangle$ ); the v20 denominators  $\frac{5}{4} - \cos(r\pi/3)$  are exactly the eigen-denominators  $|1 - \zeta_6^r/2|^2$  of the two family resolvents  $(\mathbf{1} \mp \frac{1}{2}\widetilde{M}_0)^{-1}$ . *Cyclotomic determinant [E]*:  $\det(\mathbf{1} - t\widetilde{M}_0) = 1 - t^3 = (1 - t)\Phi_3(t)$ , so  $\det(\mathbf{1} \mp \frac{1}{2}\widetilde{M}_0) = (\frac{7}{8}, \frac{9}{8})$  — and the lepton coefficients become resolvent determinants:  $c_e = |\mu_4|/(2\det_-) = \frac{16}{7}$ ,  $c_\mu = N_{\text{fam}}/(2\det_+) = \frac{4}{3}$ ,  $c_\tau = |\mu_4|\det_-/N_{\text{fam}} = \frac{7}{6}$ , product rule =  $|\mu_4|/\det_+ = \frac{32}{9}$ . The lepton 7 and 9 *are* the family-resolvent determinants;  $\delta = \frac{1}{2} = |(\widetilde{M}_0)_{22}|$  is supplied by the matrix itself. *Sheet-extended group [E] (audit)*:  $\langle U, \widetilde{M}_0, -\mathbf{1} \rangle$  has order  $48 = \Omega_{\text{adm}} = N_{\text{fam}} \dim S^+$  — the seed’s quartic coefficient. *Residue [C]*: the dictionary’s *values* are exact; its *address table* (which hexagon site each fermion occupies; the quark composition) remains v20’s established input / open — nothing re-fished.

**The address table: the lepton words are the compiler atoms [E]/[C]**  
 ([verification/v120\_address\_table.py])

The address table acquires its structure — on word-length integers *frozen* in v18/v20, long before today’s readings (the red-team firewall). (i) *The lepton words are the atoms [E]*:

$$(L_e, L_\mu, L_\tau) = (8, 5, 3) = (\text{rank } E_8, g_{\text{car}}, N_{\text{fam}}) = (p_0 + e_2, e_2, p_0)$$

in mass order (longest word = lightest lepton). (ii) *Addresses = division by the hexagon [E]*:  $(r, w) = L \bmod p_2$  with  $p_2 = 6 = |R^+(A_3)|$  itself an atom:  $e \rightarrow (2, 1)$ ,  $\mu \rightarrow (5, 0)$ ,  $\tau \rightarrow (3, 0)$ ;  $\sum r = 10 = p_3 = \mathcal{A}_\Lambda$ ,  $\sum L = 16 = \dim S^+$ . (iii) *Sheet parity [E]*: by the v118 dictionary,  $e$  sits on the *untwisted* sheet ( $\omega \in \text{spec } \widetilde{M}_0$ ),  $\mu, \tau$  on the *twisted* sheet  $(-\omega, -1)$ ; the anchor lepton  $\tau$  occupies the unique *real* twisted eigenvalue  $-1$  — exactly the site where v20’s product rule replaces the resolvent. (iv) *Quark sum rules [E] (frozen v18 words 7, 7, 5, 3, 2, 0)*: up-type sum =  $10 = p_3$ ; down-type sum =  $14 = p_1 + p_3$ ; **all quarks** =  $24 = |W(A_3)|$  (the monodromy group order, v117); **all nine charged fermions** =  $40 = p_1 p_3 = a^T R a$

(v119); quark  $\sum r = 2p_2$ ,  $\sum w = |\mathbb{Z}_2|$ ; the top sits at the vacuum site  $(0, 0)$  — zero word, no suppression, the established ladder anchor. *Residue [C]*: five independent closures  $(16, 10, 14, 24, 40)$  constrain the address map; the per-fermion *assignment* remains the v18/v20 input — deriving it is the remaining H2 content.

**The address table is pinned:  $R$  is the unique hexagon matrix with atom margins [E]**  
([verification/v121\_address\_pinning.py])

The per-fermion assignment closes at the *information* level. The established identity  $L = R + 2U$  (v95) says the word-length table *is*  $L[\text{sector}][\text{generation}]$  — rows (up; down; lepton) =  $((7, 3, 0); (7, 5, 2); (8, 5, 3))$ , the v18/v20 words verbatim — so the addresses are the entries of the *residue operator*  $R$  plus one hexagon turn for the first generation, and  $R$ 's first column is the **anchor**  $a = (1, 1, 2)$ . The margins are atoms [E]: row sums  $(4, 8, 10) = (e_1, \text{rank } E_8, p_3)$ , column sums  $(4, 13, 5) = (e_1, \Delta_Q, g_{\text{car}})$ . And the census closes it [E]:

among all  $3 \times 3$  matrices with entries in the hexagon  $\{0..5\}$ ,  
exactly 17 have the atom margins — and exactly ONE has  $\det = 8$

— and it is  $R$  (also the unique one with  $\text{SNF} = (1, 1, 8)$ ; all 17 determinants are distinct, 8 occurs once — control computed explicitly). **The assignment table carries zero residual information beyond the atom margins and the rank determinant.** Within-sector ordering is the established transport reading ( $m \sim \lambda_V^L$ : longer word = lighter fermion). *Residue [C]*: the open H2 item shrinks from “derive 9 per-fermion integers” to “derive the six atom margins +  $\det = \text{rank}$ ”. *Firewall*:  $R$ ,  $L$  and all word lengths were frozen in v4/v18/v20/v71 long before this census — no degrees of freedom spent.

**The margins are theorems** ([verification/v122\_margin\_theorem.py]). Even the margins are not input. Three selectors, all established and frozen *before* the address question was posed — (S1) the  $D_4$  annihilator  $n = (5, -9, 6)$  with  $n^T R = (8, 0, 0)$  (it kills the unchanged  $(c_2, c_3)$  plane;  $n \cdot a = 8$ ), (S2)  $\det R = 8$ , (S3)  $\det K = 4$  with  $K = R + Q\Sigma$  — pin  $R$  *uniquely* among hexagon matrices [E]: the (S1) census has  $4 \times 4 \times 4 = 64$  candidates, (S2) leaves 12, (S3) leaves **exactly one**, and it is  $R$ . Hence the atom margins, the anchor column  $Re_1 = a$  and all sum rules above are *consequences* — their input status is revoked. *Bonus lemma [E]*: on the (S1) + (S2) locus  $\det(M + 2U) = 20 = \det L$  holds for *all* 12 candidates — the fourth quartet determinant is redundant, the compiler's familiar overdetermination. *Residue [C]*: H2 reduces to the establishment of the selectors themselves ( $n$  and the diamond determinants — pre-existing inventoried objects); the address question introduces *no new residual class*.

**The dual-normal selector:**  $(d, n)$  **pins**  $R$  **columnwise**  
([verification/v136\_dual\_normal\_selector.py]). The selector list shrinks once more. With the dual anchor  $d = a^\top R^{-1}$  ([verification/v134\_dual\_anchor.py]) and the torsion normal  $n$ , each *column* of  $R$  obeys  $d \cdot c_j = a_j$  and  $n \cdot c_j = 8\delta_{1j}$ ; the common kernel of  $(d, n)$  is the primitive vector  $(6, 8, 7)$  (audit:  $p_2$ , rank  $E_8$ , scalaron-7), so within the address box  $\{0..5\}^3$  each column is the *unique* lattice point (census: 1 of 216 each) —  $(d, n) \Rightarrow R$ , and the  $6^9$  census above collapses to three column problems [E]. *Honesty*:  $K$  is *not* pinned directly ( $d^\top K = (1, \frac{1}{2}, 1)$ );  $\det K = 4$  comes via the  $\mu_4$  wall ([verification/v135\_det\_surface.py]). *The mechanism is bounded*: the zero mode of  $A_0^*$  carries the exact spectral normal  $\sigma = (2, -9, 5) = (|\mathbb{Z}_2|, -N_{\text{fam}}^2, g_{\text{car}})$  (signed squares;  $\|v\|^2 = \frac{16}{9}$  with  $16 = \dim S^+$ ,  $\text{adj } A_0^* = \frac{2}{9}P_v$  with  $\frac{2}{9}$  = the SdS entropy curvature),  $\sigma \cdot \mathbf{1} = -|\mathbb{Z}_2|$ ,  $\sigma \cdot a = N_{\text{fam}}$ ,  $n \cdot \sigma = 121 = \|\text{Pl}(K)\|_1^2$ ; and the exact  $W(A_3)$  orbit control shows *no* group image of  $\sigma$  is proportional to  $n$  — the naive cofactor/orbit lift fails, the step  $\sigma \rightarrow n$  is a genuine mechanism beyond the monodromy group.  $R_4'$  *restated [C]*: derive the *pair*  $(d, n)$ ; then  $R$  is forced columnwise.

**The selector triangle** ([verification/v139\_selector\_triangle.py]). The pair itself now collapses further. First,  $d$  is *pure anchor data*:  $d = \frac{3}{2}a - 2 \cdot \mathbf{1}$  exactly — the closed form never mentions  $R$ , so the first selector is *derived*, not residual. Second, the frame  $(\mathbf{1}, a, \sigma)$  is a basis of  $\mathbb{Z}^3$  with  $\det(\mathbf{1}|a|\sigma) = 11 = \|\text{Pl}(K)\|_1$  (the quark constant as the frame volume), and  $n$  is the *unique* integer covector with the three atom pairings

$$n \cdot \mathbf{1} = 2 = |\mathbb{Z}_2|, \quad n \cdot a = 8 = \text{rank } E_8, \quad n \cdot \sigma = 121 = \|\text{Pl}(K)\|_1^2$$

— the cofactor definition is replaced by frame data plus three atoms. On the  $(\mathbf{1}, a)$ -constrained line the  $\sigma$ -pairing is  $11(11+t)$ : divisible by 11 for *every*  $t$ , a perfect square exactly at the true  $n$  ( $t=0$ ). The



review's predicted lift is exact as an audit identity:  $n = \text{reverse}(\sigma) + |\mu_4|e_3$  (generation flip +  $\mu_4$  lift at the third slot).

**Frame integrality: three pairings → two** ([`verification/v142_frame_integrity.py`]). The third pairing is *not* independent. The pairing triples  $(m \cdot \mathbf{1}, m \cdot a, m \cdot \sigma)$  of *integer* covectors form an index-11 sublattice of  $\mathbb{Z}^3$ , cut out by the single congruence

$$x - 3y + z \equiv 0 \pmod{11} \quad (\text{index } 11 = \|\text{Pl}(K)\|_1 = \det(\mathbf{1}|a|\sigma)),$$

so with the two line pairings at their atoms  $(x, y) = (|\mathbb{Z}_2|, \text{rank } E_8) = (2, 8)$ , integrality *forces*  $z \equiv 0 \pmod{11}$  — the  $11(11+t)$  line *is* the integer line. Primitivity forces  $t$  *even* (odd  $t$  makes every entry of  $n(t)$  even), and  $t=0$  is the unique even  $t$  with a square  $\sigma$ -pairing for  $|t| < 88$  (the global pick keeps the square selector — recorded honestly). Both line pairings are Cramer-reducible to inventoried identities ( $n \cdot a = \det R$  by Laplace along the anchor column;  $n \cdot \mathbf{1} = 8(R^{-1}\mathbf{1})_1$  with  $R^{-1}\mathbf{1} = \frac{1}{4}(1, 1, -1)$ , `v134`) — restructuring, not establishment.

**The pairing values are atom identities** ([`verification/v145_pairing_atoms.py`]). Even the two line-pairing *values* carry no independent information. Reading the `v139` lift structurally —  $n = w_0(\sigma) + |\mu_4|e_3$  with  $w_0$  the  $A_3$  *exponent duality*  $m \leftrightarrow h-m$  (an established Lie involution; lift slot = the top-exponent line, lift size = the glue order) — all three pairings become atom arithmetic:

$$n \cdot \mathbf{1} = |\mu_4| - |\mathbb{Z}_2| = |\mathbb{Z}_2|, \quad n \cdot a = |\mu_4| a_3 = 8 = \text{rank } E_8 \quad \text{via} \quad |\mu_4| + g_{\text{car}} = N_{\text{fam}}^2 \quad (4 + 5 = 9, \sigma \perp w_0(a)),$$

and  $n \cdot \sigma = \|\sigma\|^2 - N_{\text{fam}}^2 + |\mu_4|g_{\text{car}} = 110 - 9 + 20 = 121$  with  $\|\sigma\|^2 = |\mathbb{Z}_2|^2 + N_{\text{fam}}^4 + g_{\text{car}}^2 = 110 = |\mathbb{Z}_2|g_{\text{car}}\|\text{Pl}(K)\|_1$ . *R4' residue [C] (final form)*: derive the *one* map  $n = w_0(\sigma) + |\mu_4|e_3$  — no independent number remains; everything else in the chain anchor  $\rightarrow (d, n) \rightarrow R$  is exact.

**Both dual normals are anchor data** ([`verification/v149_cusp_normal.py`]). The remaining opacity dissolves in the canonical frame: the pairings of  $n$  with the *cusp eigenbasis* (the geometric basis of `v98`) are exactly

$$n \cdot e_1 = 6 = p_2(a), \quad n \cdot e_2 = 3 = p_0(a), \quad n \cdot e_3 = 5 = e_2(a)$$

— one anchor invariant per  $Q_+$  degree, and the cusp matrix is unimodular, so these three pairings determine  $n$  uniquely. With  $d = \frac{3}{2}a - 2 \cdot \mathbf{1}$  the dual normal *pair* (`v134`) is anchor data through and through — one normal per frame; the cusp data  $(6, 3, 5)$ , the frame pairings  $(2, 8, 121)$  and the  $w_0$ -lift are three presentations of one covector, all atomic (equivalences exact). *Audit*:  $\sigma \rightarrow (5, 1, 2)$ ; alternate reading  $(6, 3, 5) = (|R^+(A_3)|, N_{\text{fam}}, g_{\text{car}})$ ; sum  $14 = \dim G_2$ . *R4' residue [C] (cusp form)*: derive the *assignment* — why the  $Q_+$  degrees  $(1, 2, 3)$  carry  $(p_2, p_0, e_2)$  in that order; one discrete assignment, zero continuous freedom.

### Exterior Leg Lemma — the quark $u/d$ leg is a $\Lambda^2 F$ area, not a scalar [E]/[C]

The `v41` negative is a *category error*, not a dead end.  $u$  and  $d$  share the hexagon-radial data  $(r, w) = (1, 1)$  ( $L = 7$  for both), and the  $C_6$  resolvent  $c(r, w) = |\mu_4|^w / (\frac{5}{4} - \cos \frac{r\pi}{3})$  reads *only*  $(r, w)$  — so any *scalar* leg gives  $c_u/c_d \equiv 1$ . A scalar sees radius, not orientation. The  $u/d$  separation lives in the smallest *non-scalar* invariant of the  $SU(3)_F$  composition: the exterior square  $\Lambda^2 F$ , i.e. the oriented anchor-plane area  $\text{Pl}_{1,a}(K)$ . The “11” is then *generated*, not read:

$$(K, \mathbf{1}, a) \rightarrow B_K = \begin{pmatrix} 13 & 8 & 4 \\ 18 & 11 & 6 \end{pmatrix} \rightarrow \text{Pl}(K) = (-1, 6, 4) = (-N_\Phi, |R^+(A_3)|, |\mu_4|) \rightarrow \|\text{Pl}(K)\|_1 = 11,$$

$$\frac{c_u}{c_d} = \frac{g_{\text{car}} \|\text{Pl}_{1,a}(K)\|_1}{N_{\text{fam}}^2 \Delta_Q} = \frac{5 \cdot 11}{9 \cdot 13} = \frac{55}{117} \quad (\text{anchor microcode} = \frac{e_2(a)(p_3(a)+1)}{p_0(a)^2(2p_2(a)+1)}).$$

**Typing**: [E] for the Plücker / microcode identities ( $\text{Pl}(K) = (-1, 6, 4)$ ,  $\|\cdot\|_1 = 11$ ,  $5 \cdot 11 / (9 \cdot 13) = 55/117$ ); and — by Readout Rigidity (`v49`, below) — [E] also for the *physical* ratio  $C_{ud}(\rho^*)$  = this  $\Lambda^2 F$  readout, which is *constant* on the derived selector stratum and so does *not* need any point-selection on  $\Lambda^2 F$ . The quark residual is thereby re-typed from “find the right scalar  $y$ ” to a closed exterior readout for the *ratio*; only the *absolute* amplitude scale ( $U_{\text{point}}$ ) stays as an anchor, no new scalar, no fishing.

*Audit (the 121 lemma, [verification/v119\_review\_validation\_2.py])*: the “11” has a second, independent appearance [E]: with  $h = \mathbf{1} + a = (2, 2, 3)$  in the canonical anchor ordering,  $h^T R h = 121 = 11^2 = \|\text{Pl}(K)\|_1^2 = (p_3 + 1)^2$  — the quadratic *anchor norm* of the residue operator  $R$  reproduces the Plücker integer (sensitivity control: the three orderings of  $h$  give  $\{105, 121, 135\}$ ; only



the canonical one yields 121). Bonus checksums:  $\mathbf{1}^T R \mathbf{1} = 22 = 2 \times 11$  (the entry sum of  $R$ ) and  $a^T R a = 40 = p_1 p_3 = 10 b_1 - 1$ . Typed *audit*, not load-bearing — the oriented Plücker mechanism above remains the derivation. [\[verification/v42\\_exterior\\_leg.py\]](#)

**Bridge test** ([\[verification/v43\\_exterior\\_bridge.py\]](#)). The typing is confirmed: for the  $SU(3)$  holonomy the exterior square is the conjugate representation,  $\Lambda^2(\widetilde{M}) = \widetilde{M}$  exactly — the exterior leg *is* the  $\bar{\mathbf{3}}$ . But the *continuous* action  $\Lambda^2(\widetilde{M}) \cdot (\mathbf{1} \wedge a)$  does *not* equal the *integer*  $\text{Pl}(K) = (-1, 6, 4)$ : the integer Plücker is the combinatorial  $K$  datum. So the bridge  $\rho^* \rightarrow \text{Pl}(K)$  is the *discrete* non-abelian-Hodge invariant (the integer  $R$ ;  $\det R = 8$  and  $\text{Spec}(Q_+) = \{1, 2, 3\}$  already confirmed on the bundle, Lemma U4 box), not a continuous exterior-square computation — exactly the H2 equivalence, now pinned from the  $\Lambda^2 F$  side.

**Lie-level grounding — no special math** ([\[verification/v44\\_carrier\\_exterior.py\]](#)). The discrete invariant has a home already in TFPT: the carrier half-spinor is the *even exterior algebra* of the 5-slot carrier,  $16 = \Lambda^{\text{even}}(5) = \binom{5}{0} + \binom{5}{2} + \binom{5}{4} = 1 + 10 + 5$ , and under  $SU(5) \subset SO(10)$  the up quark sits in  $10 = \Lambda^2(5)$  while the down sits in  $\bar{5} = \Lambda^4(5)$ . So the *exterior degrees* are  $\deg(u^c) = 2$ ,  $\deg(d^c) = 4$ , with  $\deg(u) + \deg(d) = 6 = |R^+(A_3)|$  (the hexagon) and  $\deg(d) - \deg(u) = 2 = |\mathbb{Z}_2|$  (the sheet). The “ $\Lambda^2$ ” of the exterior leg is therefore not exotic Hodge data — it is the carrier’s own  $\Lambda^2$  grading (the up quark *lives* in  $\Lambda^2(5)$ ). This re-types the residual once more: from “discrete non-abelian-Hodge invariant” to “the carrier exterior degree” — a standard [E] fact. *Honest scope*: this grounds the *type*; the Plücker *number* 55/117 stays the family  $\text{Pl}(K)$  [E], and the carrier- $\Lambda^2 \leftrightarrow$  family- $\Lambda^2 F$  link (via family  $\otimes$  carrier = 15 = 16–1) plus the normalisation remain [C].

**The “11” is Pascal — a third, simplest origin** ([\[verification/v45\\_family\\_exterior.py\]](#)). Following that link,  $\text{Pl}(K)$  is the *exterior algebra of the family fundamental*  $4 = \mu_4$  (the  $A_3 = SU(4)$  fundamental):  $\Lambda^k(4) = \binom{4}{k}$  is Pascal row 4 = (1, 4, 6, 4, 1) with full sum  $2^4 = 16 = \dim S^+$ , and  $|\text{Pl}(K)| = (1, 4, 6) = \{\Lambda^0, \Lambda^1, \Lambda^2\}(4)$  are exactly its distinct exterior powers. Hence

$$\|\text{Pl}(K)\|_1 = \sum_{k=0}^2 \binom{4}{k} = 11 = 2^4 - \binom{4}{3} - \binom{4}{4} = 16 - g_{\text{car}}$$

(the cumulative  $\Lambda^\bullet(\mu_4)$  up to degree 2 =  $\deg(u^c)$ ), and family  $\otimes$  carrier = 15 =  $\dim \mathfrak{su}(4) = \Lambda^2(5) \oplus \Lambda^4(5) = 10 + 5 = N_{\text{fam}} g_{\text{car}}$  read three ways. So the “11” is generated a *third* way — the cumulative family exterior algebra up to the up-quark’s degree — pure Pascal/branching, *no special math*. The [C] physical normalisation is unchanged; the algebraic origin of 55/117 is now triply grounded.

**Lemma 6** (U5 — the “11” worry, *discharged* by Readout Rigidity). *The original worry was that  $c_u/c_d = C_{ud}(\rho^*)$  must be evaluated at the uniquely selected  $\rho^*$  without using 11 as input, so that until then 55/117 would be a mere table-level rational shadow. **This is now resolved (v49):**  $C_{ud}$  is constant on the discrete selector stratum  $S_{8,Q}$ , so the “11” is earned as the rigid Plücker readout  $\|\text{Pl}(K)\|_1$  (v42) — independently of point-uniqueness. Point-uniqueness of  $\rho^*$  enters only the absolute amplitude scale, not the ratio.*

**Lemma 7** (U6 — the four certificates).  $C_U^{(1)}$  wall enumeration (done);  $C_U^{(2)}$   $D_4$ -fixed character solver;  $C_U^{(3)}$  selector uniqueness (one  $S$ -equivalence class);  $C_U^{(4)}$  amplitude extraction — the ratios  $R, Q, c_u/c_d$  are read off the (derived) selector stratum (closed, v49/v71); only the absolute  $U_f^*, c_q$  normalisation is the anchor.

**Progress: an explicit valid flat bundle is realised (RH solve)** ([E]). A numerical Riemann–Hilbert solve produced an *explicit* Hermitian residue  $A_0$  (eigenvalues  $\{0, \frac{1}{3}, \frac{2}{3}\}$ ,  $\text{diag} = (\frac{1}{2}, \frac{1}{4}, \frac{1}{4})$ ) for which the Fuchsian system  $A(z) = \sum_k U^{k-1} A_0 U^{-(k-1)} / (z - i^k)$  simultaneously satisfies: the cusp class at each puncture; the splitting  $\mathcal{O}(-2) \oplus \mathcal{O}(-1)^2$  (exponents  $\sum_k A_k = \text{diag}(2, 1, 1)$ ); the *path-ordered* monodromy at infinity is trivial,  $\|M_\infty - \mathbf{1}\| \sim 10^{-9}$ , i.e.  $M_1 M_2 M_3 M_4 = \mathbf{1}$ ; and the representation is *irreducible* — **case A** (non-trivial  $SU(3)_F$  mixing, not the diagonal case B). So a valid ( $U_{\text{wall}}$ ) flat bundle *exists and is constructed*; the U2 kill switch lands on A. *Honest scope*: this is one realising point (existence + case A), likely one of a residual family; the unique  $\nabla_F^*$  still needs the  $\det R = 8$  selector and  $c_u/c_d$  still needs the  $C_6 \leftrightarrow \mathbb{P}^1 \setminus \mu_4$  dictionary (H2). [\[verification/v33\\_explicit\\_flat\\_bundle.py\]](#)

*H2 bridge probed* ([\[verification/v34\\_h2\\_bridge\\_attempt.py\]](#)). The explicit per-puncture monodromies

$M_k$  (cusp class,  $\prod M_k = \mathbf{1}$  to  $10^{-5}$ ) were computed; but the natural resolvent-dressed diagonal extraction ( $|\text{diag } M_k| = (0, \frac{1}{2}, \frac{1}{2})$  times the  $C_6$  resolvent factor) does *not* reproduce the lepton amplitudes (0.475, 1.107, 0.917). So the precise  $\Gamma_{ij}^{\min}$  geodesic-to-word dictionary (how the 6-site hypercharge hexagon combines with the 3-dim family  $\rho^*$ ) was the genuine remaining analytic input — it is *not* fixed by a natural guess, and we do *not* fish for 55/117. *Resolved since (v118): the correct extraction is the resolvent determinant of the  $W(A_3)$  monodromy, not the diagonal dressing — the probe's failure is thereby explained, and the dictionary is exact.*

### Theorem U — the four-way split (sharper than one monster gate) [E]/[O]

$D_4$  fixes the admissible chamber, not the physical point (the  $D_4$ -fixed locus is positive-dimensional, U3); the discrete selectors  $\det R = 8$ ,  $\text{SNF}(R) = (1, 1, 8)$ ,  $\text{Spec}(Q_+) = \{1, 2, 3\}$  do the cutting. The gate splits into four pieces, three of them essentially closed:

- $U_{\text{unitary}}$  [E]: parabolic degree 0 + polystable  $\Rightarrow$  unitary (Mehta–Seshadri)  $\Rightarrow \Phi = 0$ ; the harmonic metric is the *finite* invariant form  $H$  ( $M_k^\dagger H M_k = H$ ), not a Hitchin PDE ([verification/v40\_harmonic\_metric.py]).
- $U_{H_2}$  [E]:  $R, Q, K$  are read off the bundle (splitting type =  $a$ ,  $\det R = 8$ ,  $\text{Spec}(Q_+)$ ) ([verification/v39\_uwall\_selectors.py]).
- $U_{\Lambda^2}$  [E]: **Readout Rigidity** — on the discrete stratum  $\mathcal{S}_{8,Q} = \{\det R = 8, \text{SNF}(R) = (1, 1, 8), \text{Spec}(Q_+) = \{1, 2, 3\}\}$  the anchor-plane readout is *constant*,  $\mathcal{C}_{ud}(\rho) = g_{\text{car}} \|\text{Pl}_{1,a}(K)\|_1 / (N_{\text{fam}}^2 \Delta_Q) = \frac{55}{117}$ , independent of the continuous  $D_4$  position — so  $c_u/c_d$  needs no point uniqueness ([verification/v49\_readout\_rigidity.py], [verification/v42\_exterior\_leg.py]).
- $U_{\text{point}}$  [O]: full uniqueness of  $\rho^*$  — needed only for the *complete*  $U_f^*$  amplitude matrix, *not* for  $c_u/c_d$ .

**Headline:** the remaining flavor bridge is not a PDE gate — it is a finite exterior *readout-rigidity* theorem; only  $U_{\text{point}}$  stays open.

**The simple closure** ([verification/v71\_simple\_r\_bridge.py]). The selector stratum  $\mathcal{S}_{8,Q}$  is now *fully derived*:  $\det R = 8 = \text{rank } E_8$  with  $\text{SNF}(R) = (1, 1, 8)$  is the lattice invariant (cf.  $\det Q = 3 = N_{\text{fam}}$ , v70), and  $\text{Spec}(Q_+) = \{1, 2, 3\}$  is the  $D_4$ -equivariant result (v69). The four flavor operators are integer lattice maps with compiler-atom determinants ( $\det Q, \det K, \det R, \det L$ ) = (3, 4, 8, 20), product 1920 =  $|W(D_5)|$ . So by Readout Rigidity the quark *ratios* ( $c_u/c_d = g_{\text{car}} \cdot 11 / (N_{\text{fam}}^2 \cdot 13) = \frac{55}{117}$ , etc.) are *pure integer Plücker readouts* — *no transcendental monodromy solve*. The earlier monodromy machinery was over-engineering for the ratios; the only transcendental piece,  $U_{\text{point}}$ , is the *absolute* amplitude normalisation — an anchor of the same nature as the one dimensional scale, not a missing computation.

**Gate 1 complete:**  $U_{\text{point}} \rightarrow v_{\text{geo}}$  ([verification/v75\_u\_point\_to\_vgeo.py]). The absolute amplitudes are not free either. The lepton amplitudes are exact ( $c = (\frac{16}{7}, \frac{4}{3}, \frac{7}{6})$ , v20), every cross-sector  $c$ -ratio is closed (Plücker), and each *sector product* is a clean  $\varphi_0^{\text{ret}}$ -power (Grand Mass Volume, v46:  $\det M_{\text{sector}} \sim (\varphi_0^{\text{ret}})^{6,9,10}$ ; the lepton product is  $\frac{16}{7} \cdot \frac{4}{3} \cdot \frac{7}{6} = \frac{32}{9} = 2^{g_{\text{car}}} / N_{\text{fam}}^2$ ). Since (pairwise ratios) + (product)  $\Leftrightarrow$  (individuals) is a bijection, the nine charged-fermion amplitudes are *fixed up to one overall scale*  $v_{\text{geo}}$ . **So  $U_{\text{point}}$  is not an independent gate: it reduces to  $v_{\text{geo}}$ , the same one irreducible dimensional anchor as gravity's  $1/G$  (v68) — the two [O] anchors collapse to one.**

## Research Contract 2 — ( $G_{\text{metric}}$ )

**Goal.** Construct the TFPT metric measure

$$Z[J] = \lim_{\chi \rightarrow \infty} \int_{\mathcal{G}_\chi / \text{Diff}} \exp(-S_{\text{rel},\chi}[g, A, \psi] + J \cdot O) d\mu_{\text{Cal},\chi},$$

$$S_{\text{rel},\chi} = \text{Tr} f\left(\frac{D_{\text{rel}} + A_\Sigma}{\chi}\right) - \text{Tr} f\left(\frac{D_{\text{ref}}}{\chi}\right) + \frac{i\pi}{2} \Delta\eta_\Sigma,$$

the relative spectral action whose low-curvature shadow is the already-closed  $R + R^2$  equation.

**Lemma 8** (G0–G1 — configuration space & BRST/Hodge Fredholm). *For  $s > 5/2$  a Sobolev space of  $\Theta$ -symmetric metrics on the doubled seam collar  $M_\Sigma$  admits a gauge slice; the gauge-fixed gravitational complex  $\Omega^0(TM) \xrightarrow{K} \Gamma(S^2 T^*M) \xrightarrow{K^*} \Omega^0(TM)$  with  $\Delta_{\text{gh}} = K^* K$  and  $\Pi_{\text{phys}} =$*

$\Pi_{\text{BRST/Hodge}}\Pi_{\text{TT}}$  is elliptic Fredholm; the conformal mode is removed as BRST-exact / non-propagating in the Calderón polarisation (not rhetorically). Machine: principal symbols in xAct/-Cadabra; linear-algebra certificates in Lean (medium).

**Lemma 9** (G2 — finite relative spectral action, *heat-kernel grounded* [E]).  $S_{\text{rel},\chi}$  is finite, differentiable, with a local Seeley–DeWitt expansion  $\text{Tr } f(D^2/\chi^2) \sim f_4\chi^4 a_0 + f_2\chi^2 a_2 + f_0 a_4 + \dots$ . For the Lichnerowicz Dirac operator  $D^2 = \nabla^2 + \frac{R}{4}$  (so  $E = -\frac{R}{4}$ , spinor trace  $\text{tr } \mathbf{1} = 4$ ) the Gilkey coefficients give, per  $(4\pi)^{-2}$ ,

$$a_2 = \frac{1}{6} \text{tr}(6E + R \mathbf{1}) = -\frac{R}{3} \quad (\text{Einstein–Hilbert } \int R), \quad a_4|_{R^2} = \frac{1}{360} \text{tr}(180E^2 + 60RE + 5R^2 \mathbf{1}) = \frac{R^2}{72}$$

(the  $R^2$ /Starobinsky term), so the spectral action structurally produces  $S_{\text{eff}} = \int \sqrt{g} \frac{\bar{M}_{\text{Pl}}^2}{2} (R + \frac{R^2}{6\bar{M}_{\text{Pl}}^2}) + \dots$  with the scalaron mass ratio fixed by the cutoff moment,  $M^2/\bar{M}_{\text{Pl}}^2 = 6(4\pi)^2/f_0$ . The TFPT closure  $M^2/\bar{M}_{\text{Pl}}^2 = c_3^7$  fixes  $f_0 = 6(4\pi)^2/c_3^7$ , i.e.  $M = c_3^{7/2} \bar{M}_{\text{Pl}} = 3.06 \times 10^{13} \text{ GeV}$  — the boundary normalisation  $c_3$  is the gravitational scale — and hence the Starobinsky slow-roll forms  $n_s \simeq 1 - \frac{2}{N}$ ,  $r \simeq \frac{12}{N^2}$ . The  $R + R^2$  structure is convention-independent (standard Gilkey); the precise rational  $\frac{1}{72}$  and factor 6 are the standard Dirac conventions used here ([verification/v36\_spectral\_action\_g2.py], [verification/v28\_gravity\_fR.py]).

**The simpler resolution: “full QG” for physics is the gap-decoupled admissible sector** ([E]). The infinite-dimensional ambient measure (G6) is the summit, but the admissible IR sector does not wait for it. The metric coupling is gap-dominated (G5):  $2\|V_{\text{metric}}\| = 0.785 < \Delta = 6 \log \frac{3}{2} = 2.433$ , so the admissible sector keeps a *strictly positive* effective gap  $\Delta_{\text{eff}} = 1.648 > 0$  under the stated RP and metric-coupling assumptions. A positive gap dynamically *isolates* the admissible sector from the un-built ambient/deep-UV: under those assumptions the admissible (IR) measure is closed, and the ambient completion is decoupled from it (we do *not* assert it is physically irrelevant — that would itself be a physical interpretation). So the honest reading is

|                                                                                                          |
|----------------------------------------------------------------------------------------------------------|
| admissible IR sector = closed under stated RP;<br>ambient metric measure (G6) = $G_{\text{metric}}$ gate |
|----------------------------------------------------------------------------------------------------------|

G2 supplies the local action ( $R + R^2$ , heat-kernel); G5 supplies the decoupling of the admissible sector; together they reduce ( $G_{\text{metric}}$ ) to the ambient projective limit (G6). **Precise status (Alessandro’s boundary):** the admissible IR sector is closed under the stated RP and gap-dominance assumptions; the ambient metric projective measure remains a *strict TOE completion target*. Its absence does *not* affect the bounded IR claim, but it *does block certification as a strict physical TOE* — so we do not call it “mere mathematical completeness”. [verification/v36\_spectral\_action\_g2.py]

**The glue Q-system: the index-4 object of the  $G_{\text{net}}$  statement exists explicitly** ([E]/[C]). The  $G_{\text{net}}$  target — “the seam–Calderón inclusion has Jones index  $4 = |\mu_4|$ ” — no longer refers to an unknown object. On the glue group algebra  $A = \mathbb{C}[\mathbb{Z}_4]$  (multiplication  $\delta_a \otimes \delta_b \mapsto \delta_{a+b}$ , unit  $\delta_0$ ) all Longo Q-system axioms hold exactly [E]: associativity, unit law, the Frobenius identity  $(m \otimes 1)(1 \otimes m^*) = m^*m = (1 \otimes m)(m^* \otimes 1)$ , and specialness  $mm^* = |\mathbb{Z}_4| \mathbf{1}$  — a special  $C^*$  Frobenius algebra with

|                                       |
|---------------------------------------|
| $\dim \theta = 4 =  \mu_4  = [B : A]$ |
|---------------------------------------|

exactly the Carrier Index Lemma value (KLM:  $\mu_{\text{carrier}} = 16 = 4^2 \mu_{E_8}$  re-verified). *Locality = isotropy* [E]: on the Lagrangian glue  $\langle (1, 1) \rangle$  the discriminant form gives  $q(k(1, 1)) = k^2 \in \mathbb{Z}$  for every  $k$  — the braiding obstruction vanishes (v92 in Q-system language); the two Lagrangian glues (the sheet pair) are the only maximal isotropic choices. *Halfway step* [E]:  $\{0, 2\} < \mathbb{Z}_4$  closes and gives the unique intermediate Q-system  $\mathbb{C}[\mathbb{Z}_2]$  = the  $SO(16)_1$  tower step, with index factorisation  $4 = 2 \times 2$ . *R2 sharpened* [C] (recorded, not claimed): what remains is the operator-algebraic *identification* “the seam–Calderón inclusion of the 16-Majorana net (v113) realises *this* Q-system” — matching

two constructed objects, not a search; the three loads (metric + carrier + QBL, v123) hang on this one identification.

**The hull is the graded module of the Q-system** ([verification/v128\_graded\_hull.py]). The Q-system is not merely an abstract companion of  $E_8$  —  $E_8$  *realises* it [E]: the 240 roots, built as explicit norm-2 vectors in  $D_5 \oplus A_3$ +glue coordinates, decompose as  $240 = 52 + 64 + 60 + 64$  over the glue  $\mathbb{Z}_4$  (the v89 branching as vectors), and for *all* 6720 root pairs whose sum is again a root,  $\text{coset}(r_1 + r_2) = \text{coset}(r_1) + \text{coset}(r_2) \bmod 4$ :

$E_8$  is a  $\mathbb{Z}_4$ -graded Lie algebra over the carrier; the grading group is the glue = the Q-system

— the checksum hull is the Q-system's graded module, realised in 248 dimensions. The R2 identification thus matches the Calderón inclusion against a structure that  $E_8$  already *carries*. [verification/v125\_glue\_qsystem.py]

**Graded Frobenius** ([verification/v143\_graded\_frobenius.py]). The identification holds at the finite Lie level: the graded module is in fact a graded *Frobenius* structure, i.e. everything the Q-system demands of its module is already exact inside  $E_8$  [E]: (i) *coset duality* —  $\text{coset}(-r) = -\text{coset}(r) \bmod 4$  for all 240 roots ( $-C_1 = C_3$  bijectively,  $C_0, C_2$  self-dual), so the Killing form restricts to a *nondegenerate* invariant pairing  $g_k \times g_{-k} \rightarrow \mathbb{C}$ ; (ii) the glue character average  $\frac{1}{4} \sum_k i^k \deg$  is *exactly* the carrier projector (the conditional-expectation shadow, quasi-basis index  $|\mathbb{Z}_4| = 4$ ); (iii) each glue sector is *one*  $W(D_5) \times W(A_3)$  orbit with multiplicity-one dimensions  $64 = 16 \cdot 4 = (\mathbf{16}, \bar{\mathbf{4}})$ ,  $60 = 10 \cdot 6 = (\mathbf{10}, \mathbf{6})$  — simple currents, sector ring =  $\mathbb{Z}_4$  = the  $\mathbb{C}[\mathbb{Z}_4]$  fusion, halfway subgroup  $\{0, 2\} = SO(16)_1$ . *R2 residue [C] (final form)*: exactly *one* conformal-net statement remains — the seam–Calderón inclusion of the 16-Majorana *net* carries this Q-system — with no remaining finite freedom.

**Where that statement lives: the NS/R sector census** ([verification/v148\_fock\_census.py]). The remaining work is scoped exactly. Every *untwisted* (Neveu–Schwarz) state of the 16 Majoranas has *integer* ( $D_5, SO(6)$ ) weights, and integer weights lie in the even discriminant classes — so the NS module supports only  $\{(0, 0), (2, 0), (0, 2), (2, 2)\}$ : **the odd glue classes  $k(1, 1)$  — the actual  $\mu_4$  simple currents — have zero NS support; they are twisted (Ramond-type) modules.** The Ramond zero-mode module ( $2^8 = 256$  ground states) carries exactly the four odd class pairs, 64 each, and splits  $128 + 128$  into the odd sectors of the *two* Lagrangian glues  $\langle(1, 1)\rangle$  and  $\langle(1, 3)\rangle$  (v92): *choosing the glue is choosing the Ramond projection* ( $128 = \text{one } SO(16) \text{ spinor chirality}$ ), and  $E_8$  assembles as  $248 = 120_{\text{NS}} + 128_{\text{R}}$ . *Consequence [C]*: the index-4 extension cannot even be *stated* on the untwisted module alone; R2's one remaining statement is irreducibly about the twisted-sector structure of the seam–Calderón net, with the sheet choice realised as the R-projection; the finite shadow is exhausted (v113/v125/v143 + this census).

**Lemma 10** (G3–G4 — reflection positivity, fixed and integrated). *For each  $\Theta$ -invariant  $g$  in the Calderón ball the physical boundary kernel  $K_g = P_{\text{adm}} \Pi_{\text{phys}} \mathcal{N}_g^{TT} \Pi_{\text{phys}} P_{\text{adm}}$  is reflection positive,  $\langle \Theta F, F \rangle_{K_g} \geq 0$  (G3); and  $d\mu_{\text{Cal}, \chi}$  is  $\Theta$ -invariant, supported where G3 holds uniformly, so  $\int \langle \Theta F, F \rangle_{K_g} d\mu_{\text{Cal}, \chi} \geq 0$  (G4). **G4 is the decisive step:** RP after integrating over dynamical metrics, not merely at fixed background.*

**Lemma 11** (G5 — gap dominance [E]).  $2\|V_{\text{metric}}\|_{\text{rel}} < \Delta = 6 \log \frac{3}{2}$ , with  $\|V_{\text{metric}}\|_{\text{rel}} \leq 248c_3^2 = \frac{31}{8\pi^2} = 0.39262$ , hence  $2\|V\| = 0.7852 < 2.4328$  and  $\Delta_{\text{eff}} \geq \Delta - 2\|V\| > 1.647 > 0$ . *The numerical inequality is trivial ([verification/v29\_research\_contract\_certs.py]); the work is the operator-norm bound (hard).*

**Sugawara reading + atomic OS moment** ([verification/v171\_os\_moment\_cluster.py]). The G5 budget is the  $E_8$  Sugawara constant:  $248c_3^2 = \frac{31}{8\pi^2}$  with  $31 = 1 + h^\vee(E_8)$  and  $c((E_8)_1) = 248/31 = 8$ , so gap dominance reads  $|R^+(A_3)| \log(N_{\text{fam}}/|Z_2|) > \frac{1+h^\vee(E_8)}{4\pi^2}$  — the  $A_3$  transfer gap beats the  $E_8$  Sugawara metric budget. Independently, the G3 reflection positivity is a *positive moment problem*:  $\text{spec}(T) = \{1, (2/3)^6, (1/3)^6\}$  makes every two-point correlator  $C(n) = A + B(64/729)^n + C(1/729)^n$



the moment sequence of a positive atomic measure, so each OS Hankel matrix factorises as a Vandermonde Gram and is positive semidefinite (a clean RP certificate; cluster constant 729/665, correlation length  $\xi = 1/(6 \log \frac{3}{2}) = 0.4111$ ).

**Theorem 1** (Decoupling theorem — physical low-energy claims do *not* need the full  $G_{\text{metric}}$ ). *If the admissible sector is reflection-positive (G4) and the metric coupling is gap-dominated,  $2\|V_{\text{metric}}\| < \Delta = 6 \log \frac{3}{2}$  (G5), then the admissible IR measure is stable under the ambient completion: a strictly positive effective gap  $\Delta_{\text{eff}} = \Delta - 2\|V\| > 0$  protects the admissible sector from the (un-built) ambient/deep-UV, so every physical low-energy read-out (masses, mixings,  $\alpha^{-1}$ , the  $R+R^2$  regime) is independent of the open ambient measure. Symbolically*

$$\text{RP} + 2\|V\| < \Delta \implies \text{admissible IR measure stable under ambient completion}.$$

Thus ( $G_{\text{metric}}$ ) is not a diffuse “full quantum gravity missing” block: it is the sharply delimited ambient projective limit (G6), a strict-TOE certification target whose closure the falsifiable IR physics does not depend on. **Typing:** the inequality is machine-checked [E]; the operator-norm bound on  $\|V_{\text{metric}}\|$  is the conditional input.

**Lemma 12** (G6–G7 — projective limit & Ward identities). *A projective system  $\mu_{\chi_n}$  with  $\pi_{m,n*}\mu_{\chi_m} = \mu_{\chi_n}$ , uniform moment bounds  $\sup_n \int \|g\|_{H^s}^p d\mu_{\chi_n} < \infty$ , tightness and regulator-independence yields  $\mu_{\text{QG}} = \lim \mu_{\chi_n}$  on  $\mathcal{G}/\text{Diff}$  (G6, the hardest step); the limit obeys the BRST Ward identities  $\langle s\mathcal{O} \rangle = 0$ , whence  $\nabla^\mu T_{\mu\nu} = 0$  and the  $\eta$  boundary phase cancels the APS anomalies (G7).*

**Gate 2 reduction:** G6 is a *boundary projective limit*, not a bulk one ([verification/v76\_gmetric\_reduction.py])

The exact decoupling margin is  $\Delta_{\text{eff}} = \Delta - 2\|V\| = 6 \log \frac{3}{2} - \frac{31}{4\pi^2} = 1.648 > 0$ , with  $\|V_{\text{metric}}\|_{\text{rel}} \leq 248 c_3^2 = \dim(E_8) c_3^2 = \frac{31}{8\pi^2} (31 = 2^{g_{\text{car}}} - 1)$ . **Holographic reduction:** because the seam is a *finite causal boundary*, the ambient *bulk* metric measure is reconstructed from a finite-codimension seam (Calderón) *boundary* measure — so G6 is a *boundary* projective limit. The conditional theorem is then

$$\text{RP}(\text{seam-Calderón boundary kernel}) + \text{tightness} \implies G_{\text{metric}} \text{ closes}$$

i.e. the dimension of the open problem drops from bulk QG to a finite seam-boundary measure. **Residual:** the constructive boundary projective limit (constructive-QFT grade) — reduced, not closed. So Gate 2 is now *two clean tiers*: Tier A (IR physics) closed under RP+gap; Tier B (strict TOE) reduced to a boundary measure, no longer a diffuse “full QG missing”. [C]/[O]

**And that boundary measure is a rigorously constructed net** ([verification/v77\_e8\_conformal\_net.py]). The level-1 central charges  $c = \dim \mathfrak{g}/(1 + h^\vee)$  give  $c(E_8)_1 = \frac{248}{31} = 8 = \text{rank } E_8$ ,  $c(D_5)_1 = \frac{45}{9} = 5$ ,  $c(A_3)_1 = \frac{15}{5} = 3$ , and  $c(D_5) + c(A_3) = 5 + 3 = 8 = c(E_8)$  is a *conformal embedding*  $(D_5)_1 \times (A_3)_1 \subset (E_8)_1$  (coset  $c = 0$ ). Now  $(E_8)_1$  is the holomorphic  $c = 8$   $E_8$ -lattice VOA / conformal net — one of the most rigorously constructed objects in mathematical QFT (Frenkel–Kac–Segal; Kawahigashi–Longo / Carpi). **So the conditional theorem becomes concrete:** *identify the seam-Calderón boundary measure with the  $(E_8)_1$  lattice net* (carrier  $= (D_5)_1 \times (A_3)_1$  subnet), and the gap  $\Delta_{\text{eff}} > 0$  supplies the clustering/tightness — so G6 is imported into existing RCFT/conformal-net rigor rather than built from scratch. Residual = the net identification + bulk reconstruction. [E] (charges, embedding) + [C] (identification)

**The Simple-Current Extension Theorem — the central  $G_{\text{net}}$  closing statement [E]/[C]** ([verification/v154\_simple\_current\_theorem.py])

The conditional theorem can now be stated as one clean *simple-current extension*. Let  $A = (D_5)_1 \otimes (A_3)_1$  be the carrier net and  $L = \langle (1, 1) \rangle \subset \text{Disc}(D_5) \oplus \text{Disc}(A_3) = \mathbb{Z}_4 \oplus \mathbb{Z}_4$  the isotropic  $\mu_4$  glue line ( $q(k(1, 1)) = k^2 \in \mathbb{Z}$ ). Then the local extension  $B = A \rtimes L$  has

$$[B : A] = |L| = 4 = |\mu_4|, \quad c(B) = 5 + 3 = 8, \quad \mu(B) = \frac{\mu(A)}{|L|^2} = \frac{16}{16} = 1$$

so  $B$  is *holomorphic* (KLM  $\mu$ -additivity), and a holomorphic  $c=8$  chiral net is the lat-



tice net of the *unique* even-unimodular rank-8 lattice — hence  $\boxed{B \cong (E_8)_1}$  (the same-*c* rival  $SO(16)_1$ ,  $\mu=4$ , is excluded). Every algebraic ingredient is machine-checked: the  $\mathbb{C}[\mathbb{Z}_4]$  Longo Q-system ([[verification/v125\\_glue\\_qsystem.py](#)]), the  $\mathbb{Z}_4$ -graded Frobenius  $E_8$  hull ([[verification/v143\\_graded\\_frobenius.py](#)]), the Ramond glue sectors ([[verification/v148\\_fock\\_census.py](#)]). **Consequence:**  $G_{\text{net}}$  is internally *closed* if the seam-Calderón boundary net is *defined* to be  $A$ ; it stays the *one* external identification “the RP seam-Calderón net is  $A$ ” if the boundary net is to be *derived* from the seam kernel. Either way this is the single strict-TOE certification theorem still carrying weight — a clean extension statement, not “find a quantum gravity”.

**And that one identification reduces to a single shared premise** ([[verification/v155\\_quasifree\\_boundary.py](#)]). The identification splits (v110) into (a) “the seam Calderón involution is sheet-odd” — the double-cover deck, the same  $|\mathbb{Z}_2|$  that halves  $c_3 = \frac{1}{2 \cdot 4\pi}$  (structural) — and (b) “the seam 2-point kernel is quasi-free”. Part (b) is *not independent*: the boundary marginal of a reflection-positive *Gaussian* bulk is the compression  $PTP$  of the bulk covariance, and the compression of a contraction ( $i\Gamma$  with spectrum in  $[-1, 1]$ ) is a contraction — so the boundary/seam state is automatically quasi-free, with Wick inherited (every boundary correlator a Pfaffian of  $\Gamma_d$ ). Hence (b) *follows* from “the seam bulk is a reflection-positive free field” — the *same* premise already load-bearing in the area law ([[verification/v59\\_area\\_law\\_evidence.py](#)]) and the EH mechanism ([[verification/v150\\_replica\\_eh\\_model.py](#)]/[[verification/v151\\_bfk\\_split.py](#)]). **So the entire seam-side residual is one physical premise** — the seam bulk is a free RP field — shared by  $G_{\text{net}}$ , the area law and R3, and given it the chain Gaussian bulk  $\Rightarrow$  quasi-free boundary  $\Rightarrow$  one kernel = net (v113)  $\Rightarrow$  carrier  $A \Rightarrow (E_8)_1$  is exact. It is not reducible by a finite computation — it is the one place TFPT meets functional analysis.

**The construction, exhibited and verified** ([[verification/v156\\_seam\\_net\\_construction.py](#)]). The “ $B \cong (E_8)_1$ ” step is no longer an assertion — the chiral net is built explicitly and the identification *checked*. The Dirichlet-to-Neumann (Calderón) operator of the 2d Laplacian acts on a boundary mode  $e^{ikx}$  by the harmonic extension  $e^{ikx-|k|y}$ :  $-\partial_n u|_0 = |k|u$ , so for a free bulk the seam operator is the massless chiral dispersion  $\omega_k = |k|$  that defines the free-fermion net on  $S^1$ . Sixteen Majoranas give  $c = 8 = 5+3$ ; the weight-1 currents are  $248 = 120_{\text{NS}} + 128_{\text{R}}$  (the  $SO(16)$  bilinears plus the Ramond spinor, Frenkel-Kac); and the holomorphic character is

$$\chi = \frac{E_4}{\eta^8} = q^{-1/3} (1 + 248q + 4124q^2 + 34752q^3 + 213126q^4 + \dots) = j^{1/3}$$

(verified by  $q$ -series), the 248 at level 1 being  $\dim E_8$  — so the net *is*  $(E_8)_1$ . *What remains* is exactly (i) the free-bulk premise above and (ii) the operator-algebraic existence / uniqueness of the net generated by the seam Calderón kernel (a Kawahigashi-Longo-type net-construction theorem). Route 3 thus *exhibits and verifies* the target; the irreducible residual is the one premise + the standard net-existence step — where TFPT meets constructive QFT, now sharply located.

**And the free-bulk premise itself reframes: freeness is a rigid fixed point, not a postulate** ([[verification/v157\\_rigid\\_fixed\\_point.py](#)]). Three facts click together. (i) The DtN/Calderón symbol is *universally*  $|k|$ : the boundary operator of *any* second-order elliptic bulk is an order-1  $\Psi\text{DO}$  with principal symbol  $|\xi|_g$  (Lee-Uhlmann), so the free chiral dispersion is bulk-detail-independent and any interaction lives in the lower-order (irrelevant) symbol. (ii) A holomorphic  $c=8$  theory is *rigid*: every field has weight  $(h, \bar{h}=0)$ , an exactly-marginal deformation needs  $(1, 1)$ , and with  $\bar{h}=0$  *no*  $(1, 1)$  field exists — the fixed point is isolated, there is no continuous knob along which a non-Gaussian interaction could turn on. (iii) The *same*  $|\mathbb{Z}_2|$  that halves the one-sided  $S^2$  Gauss-Bonnet ( $c_3 = \frac{1}{|\mathbb{Z}_2| 2\pi \chi(S^2)} = \frac{1}{8\pi}$ ) labels the sheet *and* is the Ramond/GSO projection giving  $\mu=1$  ( $248=120_{\text{NS}}+128_{\text{R}}$ ) — one object, three roles, so the geometry fixing  $c_3$  is the projection making the net holomorphic. **Consequence:** premise (A) shrinks from “prove the seam is Gaussian” to “the gapped ( $\Delta=6 \log \frac{3}{2} > 0$ ) boundary flow reaches the holomorphic fixed point” — given which, freeness is *forced* by rigidity + uniqueness ( $(E_8)_1$ , v83), not fine-tuned. A strictly smaller and simpler residual.

**And the destination is a proven stable fixed point** ([[verification/v158\\_fixed\\_point\\_stable.py](#)]). An exact operator-dimension count settles it: the chiral Majorana has  $h=\frac{1}{2}$ , the bilinear current  $h=1$ , the quartic  $h=2$ . The relevant window  $0 < h < 1$  for bosonic deformations is *empty* (bosonic chiral operators have integer  $h$ , lowest 1) — nothing relevant flows the free theory away; the  $h=1$  currents are chiral (v157), not  $(1, 1)$ -marginal; and the lowest genuinely-interacting operator (the quartic,  $h=2>1$ ) is *irrelevant*, as

are all higher  $2k$ -fermi operators ( $h=k$ ). So the free chiral  $c=8$  fixed point is *isolated and stable* against all interactions. Premise (A) thereby reduces to a *basin-of-attraction* statement with a proven stable target — the gapped ( $\Delta=6\log\frac{3}{2}$ ) flow lands here; the convergence is conformal-perturbation analysis, the destination is settled.

**And the closure can be stated as one fixed-point theorem — which relocates (A) to a single internal premise** ([`verification/v160_seam_gaussianity_from_pf.py`]). Phrase the argument internally: (i) a quasi-free state has *all* higher connected cumulants  $\kappa_{2n}=0$  (the signed set-partition/Pfaffian inversion of  $\langle\psi_1\cdots\psi_{2n}\rangle = \text{Pf}(\Gamma)$  leaves only the degree-2 kernel — the v113 “one kernel = the net” property at the cumulant level); (ii) a kernel-preserving transport fixes the whole quasi-free state by Wick functoriality; (iii) a gapped primitive Perron–Frobenius transport has a *unique* fixed ray, so a non-Gaussian piece is impossible — if  $\kappa_{2n}$  were fixed it would be a *second* fixed ray (killing uniqueness), and if not fixed it sits in the gapped complement and decays to 0 (any gap  $> 0$  suffices; the value  $(2/3)^6$  only sets the rate). *The honest residual*: v56’s gapped PF uniqueness is established on the *three-dimensional* scalar transfer spectrum  $\{1, (2/3)^6, (1/3)^6\}$  (the flavour gap / horizon Page-recovery rates), whereas the 16-Majorana seam state already carries  $\binom{16}{4}=1820$  and  $\binom{16}{6}=8008$  higher-cumulant directions — the Schwinger cone the transport must contract has dimension  $\gg 3$ , and uniqueness must hold there (not merely on the quasi-free sub-cone) for the argument to be non-circular. **So (A) is equivalent to one strictly smaller, internal, exact-testable statement**: the admissible TFPT seam transport is a *primitive, spectrally gapped, RP-cone-preserving* map on the *full* Schwinger cone, not only on the scalar readout spectrum of v56 — which the suite does *not* yet establish [E]/[O].

**And that full-cone premise is now discharged to a single finite one-particle statement** ([`verification/v161_one_particle_reduction.py`]). The lever is Bogoliubov second quantisation: a quasi-free transport is  $T_\Sigma = \Gamma(t)$ , the second quantisation of a one-particle contraction  $t$  on the 16-dimensional Majorana space  $H_\Sigma$ , and the Fock space grades by correlator degree, so  $\Gamma(t)$  acts on the degree- $m$  sector as the exterior power  $\Lambda^m(t)$ . Three exact consequences follow (all machine-checked): (i)  $\text{spec}(\Gamma(t)|_{\text{deg } m})$  is the set of products of  $m$  one-particle eigenvalues — the whole correlation spectrum is generated by the finite one-particle spectrum; (ii) **the cone gap equals the one-particle gap**: the sub-leading eigenvalue of  $\Gamma(t)$  over *every* degree is  $(2/3)^6$  (pad the largest contracted mode with fixed factors), so premise 5 on the cumulant space *is* premise 5 on the 16-dim space, the same number; (iii) the eigenvalue-1 sector is  $\Lambda^*$  of the fixed subspace — the disconnected Wick powers of  $K_\Sigma$  — so with a strict one-particle gap no independent connected cumulant is fixed, and an independent fixed cumulant *on* the fixed subspace is excluded by v113 (one polarisation  $\Leftrightarrow$  one vacuum). Premise 1 (quasi-freeness) is itself *forced*, not assumed: the  $120 = \dim \mathfrak{so}(16)$  weight-1 chiral currents *are* the Bogoliubov generators on the 16 Majoranas, and v158 makes every non-Gaussian deformation irrelevant. **So the three full-cone premises collapse to one finite, 16-dimensional statement**: the seam RG transport’s one-particle symbol is the gapped Calderón Bogoliubov map — fixed eigenspace = the rank-8  $K_\Sigma$  polarisation (v113), sub-leading gap =  $(2/3)^6$  (v56). Every *number* in it is already supplied by the suite; the one irreducible step is the physical identification that the actual seam RG generator equals this Bogoliubov map — the same operator-algebraic (Kawahigashi–Longo-class) statement as the net-existence residual of  $(A_2)$ , now with the infinite-dimensional cone eliminated [E]/[O].

**And that finite statement carries no new open content: it factors into the two already-open items** ([`verification/v162_seam_transport_identification.py`]). Take the one-particle symbol apart into its two data, the gap value ( $\beta$ ) and the fixed space ( $\alpha$ ). ( $\beta$ ) *is forced*: a  $\mu_4$ -equivariant operator is diagonal in the cusp basis — the deck average is the diagonal projection,  $\sum_{k=0}^3 U^k X U^{-k} = |\mu_4| \text{diag}(X)$  for  $U = \text{diag}(1, i, -i)$ , and the commutant of  $U$  is exactly the diagonal algebra — so its spectrum is the cusp weights  $\text{spec}(A_0^*) = \{0, \frac{1}{3}, \frac{2}{3}\}$  (the parabolic residue, v115), and the transfer eigenvalue  $(1-\alpha)^{p_2}$  with  $p_2=6=|R_+(A_3)|$  gives  $\{1, (2/3)^6, (1/3)^6\}$ , gap  $6\log\frac{3}{2}$  — every entry carrier data, no free knob; the lift to the 16 Majoranas is the established  $A_3$  subnet grading (v137). ( $\alpha$ ) is the rank-8 Calderón polarisation of the  $\omega_k=|k|$  free dispersion (v113/v156), the IR conformal free-fermion vacuum. **What is left** is only that the seam RG generator *is*  $\mu_4$ -equivariant and admissible — which is the R1 “seam clock” identification (the transfer operator as the seam’s gapped generator) — together with the  $A_2$  net-existence theorem. Both are pre-existing open items, so premise (A) introduces *no independent open content*: it factors exactly into  $A_2 + R_1$  with every number forced. The “free-bulk cloud” is gone; what remains is the same two identifications the programme already names — (A) is closed *conditionally* on them (a unification, honestly not an unconditional proof) [E]/[O].

**And both residual identifications are pinned to their floor, so (A) has no independent open gate left** ([`verification/v165_premise_a_closure.py`]).  $A_2$  (net existence) is not open research

but *assembly* of established mathematics: the 16 free Majoranas define a free-fermion conformal net (Buchholz–Mack–Todorov), the holomorphic  $c=8$  lattice net *is*  $(E_8)_1$  (Frenkel–Kac–Segal / Dong–Xu / Kawahigashi–Longo), the  $\mu_4$  extension is a Longo Q-system of index 4 (v125); the one TFPT-specific input — that the seam Calderón kernel is the free-fermion two-point function — is done because the DtN symbol is universally  $|k|$  (v156/v157). So  $A_2$  re-types **[O]**  $\rightarrow$  **[C]** (closed by assembly).  $R_1$  (the seam clock) reduces to **GATE.QGEO**: a  $\mu_4$ -equivariant flat gapped generator on the carrier is *unique* ( $= A_0^*$ , v115/v116), so its gap  $(2/3)^6$  is forced once the seam generator lies in that class, and the residual is exactly one geometric identification — the seam collar boundary is the  $\mu_4$ -punctured sphere  $\mathbb{P}^1 \setminus \mu_4$  — whose cohomological half is already established (v137,  $b_1=3=N_{\text{fam}}$ ). That is **GATE.QGEO**, an already-open realisation premise, *not* a new gate. By the No-Unit Theorem (v153) the irreducibles stay exactly  $\{\pi, v_{\text{geo}}\}$  — closing (A) adds none. **Final accounting:** premise (A) = ( $A_2$  assembly, **[C]**) + ( $R_1=\text{GATE.QGEO}$ , already open) + the  $\{\pi, v_{\text{geo}}\}$  irreducible core (a theorem) — it carries *zero* independent open gates and ceases to be its own gate **[E]/[O]**.

**And the two structural residuals — net existence and full-cone reflection positivity — are now discharged to a theorem, leaving the seam realisation as the one irreducible open premise** (`[verification/v175_net_existence_full_cone.py]`). The CAR second-quantisation functor  $\Gamma(t) = \bigoplus_m \Lambda^m(t)$  lifts the one-particle seam contraction  $t$  to the *complete* Fock space  $\Lambda^\bullet(\mathbb{R}^{16})$  ( $\dim 2^{16}=65536$ ); by the exterior-power spectrum theorem every eigenvalue of  $\Gamma(t)$  is a subset product of  $\text{spec}(t)$ , so checking all  $2^{16}$  products shows  $\Gamma(t)$  is a *contraction* (full-cone RP), with eigenvalue-1 multiplicity  $2^8=256$  (the wedges of the rank-8  $K_\Sigma$  polarisation) and sub-leading eigenvalue exactly the one-particle gap  $(2/3)^6$  — so full-cone RP/gap reduce *for every*  $m$  (not only  $m \leq 6$ , v161) to the one-particle contraction (Shale–Stinespring), *discharging the v160 scope premise by a theorem*. And  $A_2$  is now an *assembled and verified* certificate: the  $(E_8)_1$  character  $E_4/\eta^8 = 1 + 248q + 4124q^2 + 34752q^3 + 213126q^4 + 1057504q^5$  (level-1=248), the embedding  $248=120+128$ , and the  $E_8$  Cartan matrix even with  $\det 1$  (the unique rank-8 even unimodular lattice, v83) — the net *exists* by cited theorems. **Both residuals then collapse to the same input:** that the actual seam Calderón kernel is this gapped one-particle contraction on  $\mathbb{P}^1 \setminus \mu_4$ , i.e. **QGEO.REALIZE.01**, whose finite half is proven (v168). That realisation half is a constructive-geometry/AQFT statement, *not* closeable by a finite computation, and is left honestly **[O]** — the single irreducible structural premise; net existence and full-cone RP are **[E]**.

**The No-Unit Theorem —  $v_{\text{geo}}$  is an irreducible metrology primitive **[E]/[O]**** (`[verification/v153_no_unit_theorem.py]`)

$v_{\text{geo}}$  is not an open gap. Every load-bearing compiler datum —  $g_{\text{car}}, |\mu_4|, N_{\text{fam}}, \text{rank } E_8, \alpha^{-1}$ , the determinant ladder  $(3, 4, 8, 20)$ ,  $c_3 = \frac{1}{8\pi}$ ,  $\varphi_0^{\text{ret}}$  — is mass-dimension 0 and *invariant* under a unit rescaling  $L \mapsto \lambda L$ , whereas a mass / energy /  $1/G$  carries nonzero dimension. A map from dimension-0 inputs to a dimension- $d \neq 0$  output is simultaneously invariant and covariant: impossible unless the unit is introduced. **So a dimensionless boundary compiler provably cannot select an absolute scale** —  $v_{\text{geo}}$  is a *forced input*. The three apparent residuals then collapse onto the one unit, differing only in mass-dimension:

$$U_{\text{point}} \sim v_{\text{geo}}, \quad \frac{1}{G} \sim v_{\text{geo}}^2, \quad m/\mu = e^{3/4} \text{ (a pure ratio),}$$

so the irreducibles stay exactly  $\{\text{one dimensionful unit}, \pi\}$ . *Re-typing:*  $v_{\text{geo}}$  is an *irreducible metrology primitive* (a declared unit, like choosing a metre), not a missing derivation.

**Theorem G — the closure statement **[O]** (target)**

The relative spectral boundary kernel over the diffeomorphism-quotiented metric sector admits a regulator-independent, reflection-positive projective-limit measure  $\mu_{\text{QG}}$ . Its Ward identities imply covariant conservation, and its low-curvature expansion is the TFPT  $R + R^2$  seam action

$$f_R R_{\mu\nu} - \frac{1}{2} f(R) g_{\mu\nu} + (g_{\mu\nu} \square - \nabla_\mu \nabla_\nu) f_R = \bar{M}_{\text{Pl}}^{-2} T_{\mu\nu} + O(R^3/M^4),$$

so the closed covariant equation becomes the controlled IR shadow of the full measure. *Status: open; the deepest item. G5 certified; the regime equation closed* `[verification/v28_gravity_fR.py]`.

## Certifiability and recommended order

| Building block                                | Machine                 | Hardness             |
|-----------------------------------------------|-------------------------|----------------------|
| $W_{\text{wall}}$ enumeration ( $C_U^{(1)}$ ) | Lean / Python / Wolfram | easy ( <i>done</i> ) |
| $D_4$ fixed-locus polynomial ( $C_U^{(2)}$ )  | Sage / Mathematica      | medium               |
| selector uniqueness ( $C_U^{(3)}$ )           | interval arithmetic     | medium               |
| SNF, determinants, spectra                    | Lean                    | easy–medium          |
| principal-symbol ellipticity (G1)             | xAct / Cadabra / Lean   | medium               |
| gap inequality (G5)                           | Lean                    | easy ( <i>done</i> ) |
| operator-norm bound (G5)                      | analysis + numerics     | hard                 |
| projective limit (G6)                         | not primarily machine   | very hard            |
| Ward identities (G7)                          | BV/BRST formal + checks | hard                 |

### Recommended order.

 Selector-triangle pairings  $\rightarrow v_{\text{geo}}$  scale anchor  $\rightarrow G_{\text{net}}$  inclusion theorem

The flavor interface (historically  $U_{\text{wall}}$ ) is now reduced to the *selector-triangle pairings* (the dual normal pair  $(d, n)$  forcing  $R$  columnwise, v136/v139); it is finite, falsifiable and the cheapest visible upgrade. The  $v_{\text{geo}}$  scale anchor is the one dimensional input (same nature as  $1/G$ ).  $G_{\text{net}}$  — the index-4 seam-net inclusion certifying the metric sector — is more fundamental but a *programme*, not a single proof: heavy machinery, with the projective limit the genuine summit.  $F_{\text{transfer}}$  (source $\rightarrow$ pole / relic / cosmology) is the remaining downstream interface.

$F_{\text{transfer}}$  **is a typed functor, not a bag of open topics.** It is standard physics fed TFPT *source* data,  $F_{\text{transfer}} = F_{\text{observable}} \circ F_{\text{threshold}} \circ F_{\text{RG}}$ , with exactly four interfaces:

| Interface              | Map                                                                         | Status    | Source data $\rightarrow$ observable                                                                                                                                                                                                                                     |
|------------------------|-----------------------------------------------------------------------------|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $F_{\text{pole}}$      | source $\rightarrow$ pole masses                                            | [E] / [C] | Koide 53/54 = $a^T(R+Q)1/(21^T Ra)$ is an exact operator factor [E] (the missing Sheet-Diamond corner); the pole-transfer interpretation is [C] ([verification/v183_koide_f_corner_transfer.py])                                                                         |
| $F_{\text{Boltzmann}}$ | CP source $\rightarrow \eta_B$                                              | [C]       | $\tilde{m}_1=m_3/A_\Lambda$ anchors the washout, but $M_1=M_R\varphi_0^4$ only relocates the free input ( $M_R$ is the seesaw scale, not a compiler power), so $\eta_B$ stays a <i>sharper scenario</i> , not a closure ([verification/v184_etaB_anchored_boltzmann.py]) |
| $F_{\text{relic}}$     | $(f_a, m_a, \theta_i, N_{\text{DW}}) \rightarrow \Omega_a$                  | [C]       | the determinant-line axion can be all of DM, but only on the $\theta_i \approx 170^\circ$ hilltop where the relic is exponentially sensitive — a viable scenario ([verification/v185_axion_relic_solver.py])                                                             |
| $F_{\text{QCD}}$       | $(\alpha_s, b_3, \Lambda_{\text{QCD}}, \text{lattice}) \rightarrow m_p/m_e$ | [O]       | a QCD matching contract ( $b_3=-7$ a carrier output), <i>not</i> a compiler number — the rejected 1836 near-formula is the discipline working                                                                                                                            |

A machine guard enforces the typing: the four families must always carry a [C]/[O] marker and must never be promoted to a primitive [E] compiler prediction ([verification/v187\_fttransfer\_laws.py]). Exact *algebraic* sub-parts (the Koide factor,  $b_3=-7$ ) may be [E]; the *physical* prediction never is. So the visible residual is one geometric postulate (QGE0.SYM.01), one unit gauge ( $v_{\text{geo}}$ ), and this one typed transfer functor — everything else is out-of-sample data.